A cross-cohort single-cell and spatial atlas of human H3K27M-altered diffuse midline glioma

[Author One][1], [Author Two][1,2], and [Senior Author][1,*]

[1][Department, Institution, City, Country]
[2][Second affiliation]
[*]Correspondence: [email]

Keywords: diffuse midline glioma; H3K27M; single-cell RNA sequencing; single-nucleus RNA sequencing; spatial transcriptomics; data integration; reference atlas

Manuscript type: Resource. **Main figures:** 5. **Tables:** 4. **Supplementary items:** 7 figures, 1 table.

Highlights

- A harmonized reference of 311,618 cells from 57 human H3K27M DMG samples and 52 patients
- Only 2 of 15 malignant expression programmes recur across studies and both assay types
- Every lineage-identity programme is nuclei-restricted; dissociation stress is cell-restricted
- Mesenchymal tumour co-localises with myeloid cells; the OPC-oligodendroglial pairing does not

In brief

[Authors] integrate six published single-cell and single-nucleus datasets of H3K27M diffuse midline glioma into a harmonized 52-patient reference, then use it to show that most malignant expression programmes are assay-dependent rather than reproducible features of the disease. The atlas serves as a reference for mapping new DMG data.

Summary

H3K27M-altered diffuse midline glioma (DMG) is uniformly lethal, and the single-cell literature describing it is fragmented across small cohorts that differ in dissociation protocol, assay and anatomical site. We integrated six published human DMG datasets into a harmonized reference of 311,618 cells and nuclei from 57 samples and 52 patients, with lineage labels that transfer to entirely held-out patients at 0.975 accuracy and malignant calls that agree with the original annotations in a median 89.8% of cells per sample. Applying a replication rule stated before the analysis, only 2 of 15 malignant expression programmes recur across three studies and both assay types, because every lineage-identity programme in these data is nuclei-restricted. In 21 Visium sections from 12 patients, a pre-stated mesenchymal–myeloid association holds while a pre-stated oligodendroglial one does not. The atlas is released as a reference for mapping new DMG data.

Introduction

Diffuse midline glioma with an H3 K27 alteration is a distinct WHO grade 4 entity of the pons, thalamus and spinal cord, and remains incurable: median survival is under a year and no systemic therapy has altered that[1–3]. The defining lesion, a lysine-to-methionine substitution at position 27 of histone H3, was identified in paediatric glioblastoma and diffuse intrinsic pontine glioma[4,5] and acts by inhibiting PRC2, producing global loss of H3K27me3 with focal retention and widespread H3K27 acetylation[6–9]. The resulting transcriptional dependencies[10], the capacity of these tumours to integrate electrically into neural circuits[11] and their subclonal functional diversity[12] have all been mapped, yet neither targeted epigenetic therapy[13] nor cellular immunotherapy[14] has produced durable remission.

Single-cell studies established that DMG is dominated by an oligodendrocyte-precursor-like (OPC-like) compartment with smaller astrocyte-like (AC-like) and oligodendrocyte-like (OC-like) fractions, a picture first drawn from 3,321 cells across

six patients[15] and subsequently refined by developmental anchoring in canonical and non-canonical H3 variants[16,17] and by age-stratified analysis of tumour cell states[18]. The conceptual framework — a hierarchy of cycling progenitor-like cells differentiating into more mature states — parallels adult glioblastoma[19,20]. The microenvironment has been characterised as non-inflammatory and microglia-dominated[21,22], with candidate immune checkpoints emerging from recent cohorts[23], and spatial studies have begun to place these states in tissue[24,25].

Three problems limit how far that literature can be pushed. First, every individual DMG cohort is small; few public deposits contain more than ten patients, so sample-level inference is underpowered and patient-level inference rarer still. Second, the deposits differ in ways confounded with the biology of interest: pontine tumours are sampled at autopsy as frozen nuclei[26] while thalamic tumours are more often resected and dissociated as whole cells[27], so assay, site and clinical context cannot be separated within any one study. Third, published cell states are typically derived per study, so it is unclear which are reproducible features of the disease and which are properties of a protocol — a concern made concrete by pan-cancer work showing that recurrent expression programmes can be recovered consistently only when tested across many tumours[28].

An integrated reference addresses the first problem and makes the second and third testable. Variational autoencoders trained with batch covariates[29] and their semi-supervised extension for label transfer[30] are standard for atlas assembly, with transfer learning allowing new datasets to be placed in an existing reference[31] and established metrics for judging whether batch effect was removed at the cost of biology[32]. We applied this machinery to every human H3K27M DMG single-cell deposit we could obtain and treated the limitations as first-class results: we pre-registered a replication rule for expression programmes, kept patients rather than cells as the unit of inference wherever a claim concerns tumours, and submitted the finished analysis to an adversarial audit followed by independent verification of every correction.

Results

A 311,618-cell integrated reference from six human DMG cohorts

We screened the public single-cell landscape for human H3K27M DMG and retained six studies as a reference cohort, with five further cohorts reserved for projection-only validation (Supplementary Table 1). Quality control was applied per assay class, and of 420,381 cells in atlas-eligible samples 356,406 passed (84.8%); the reference was then restricted to samples with recoverable unique molecular identifier counts, giving **311,618 cells and nuclei from 57 samples and 52 patients** measured over the 18,882 genes detected in every contributing study (Table 1, Figure 1a). One autopsy series dominates: Okada 2026 contributes 156,936 nuclei from 22 patients (50.4%)[26], followed by Jessa 2022 single-cell and single-nucleus data (83,076 cells, 17 patients), the same study's multiome RNA (46,288 nuclei, 10 patients)[16], Collot 2026 (17,562 nuclei, 4 patients)[23], Zang 2024 (6,546 cells, 1 patient)[33] and DeSisto 2024 (1,210 cells, 2 patients)[27]. By assay the reference is 229,043 single-nucleus, 46,288 multiome RNA and 36,287 whole-cell profiles — 88.4% nuclei — and by site 42 pontine and 15 thalamic samples. The H3 variant is reported for only 32 of 57 samples (8 H3.1, 1 H3.2, 23 H3.3); the remaining 25 are DMG by deposited diagnosis.

Integration with scVI[29] (three seeds, 3,000 highly variable genes selected per sample[34], sample as batch, study, assay and mitochondrial fraction as covariates) was compared against Harmony[35] and uncorrected principal components using scIB-style metrics[32]. The pre-stated selection score — 0.4× batch mixing + 0.6× biological conservation — chose scVI seed 1 at **0.739** (batch 0.537, biology 0.874) over Harmony at 0.581 (0.736, 0.478) and principal components at 0.126 (0.127, 0.125) (Figure 1e, Supplementary Figure 2). Harmony mixes batches more aggressively but at a cost in biological structure that the weighting rejects. In the chosen embedding, malignant cells form a single large territory while myeloid, oligodendrocyte, astrocyte, neuronal, vascular and T/NK compartments separate cleanly, and samples from all six studies and all three assays intermingle within each compartment (Figure 1b–d).

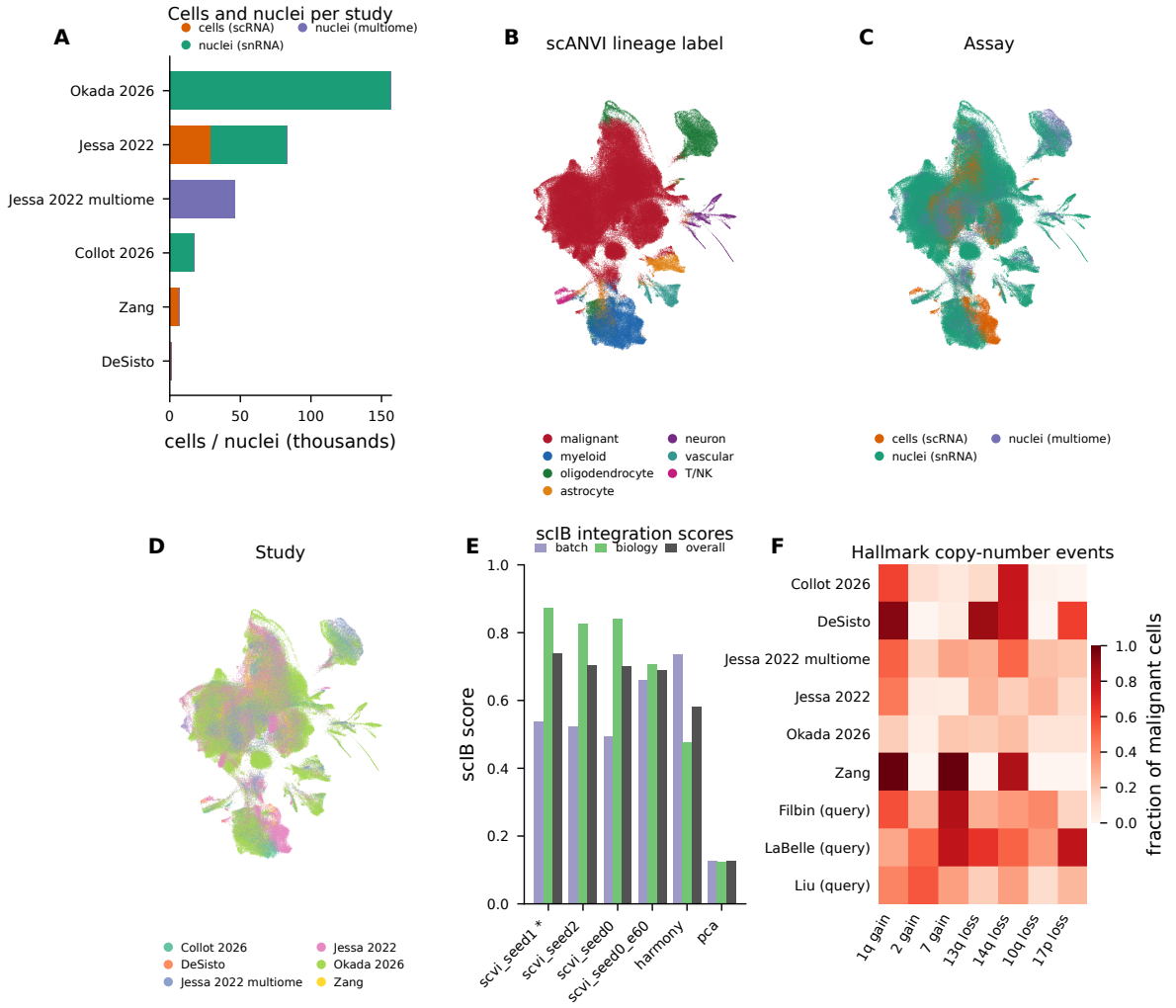


Figure 1: An integrated reference of human H3K27M diffuse midline glioma. **a**, Cells and nuclei per contributing study, coloured by assay. **b–d**, UMAP of the scANVI latent space (computed on "X_scanvi" with 30 neighbours; 311,618 cells, 57 samples, 52 patients), coloured by harmonised lineage label (b), assay (c) and study (d). **e**, scIB-style integration metrics for three scVI seeds, a 60-epoch scVI variant, Harmony and uncorrected principal components, separated into batch-mixing, biological-conservation and overall scores; the asterisk marks the model selected by the pre-stated 0.4/0.6 score. **f**, Fraction of copy-number-malignant cells carrying each recurrent arm-level event, by study, including the two projection-only cohorts.

Table 1: Reference cohort. Samples, patients and cells per contributing study after quality control, with assay, anatomical site and reported H3 variant (sample counts), and the fraction of cells called malignant by the consensus rule. Counts are cells for whole-cell and nuclei for single-nucleus and multiome assays.

Study	Samples	Patients	Cells	Assay (cells)	Site (samples)	H3 variant (samples)	Malignant
Okada 2026	22	22	156,936	snRNA:156936	thalamus:6;pons:16	unknown:22	0.516
Jessa 2022 (sc/snRNA)	17	17	83,076	snRNA:54545;scRNA:28531	pons:13;thalamus:4	H3.1:6;H3.2:1;H3.3:10	0.566
Jessa 2022 (multiome)	10	10	46,288	multiome_RNA:46288	pons:9;thalamus:1	H3.1:2;H3.3:8	0.595
Collot 2026	5	4	17,562	snRNA:17562	pons:3;thalamus:2	H3.3:5	0.570
Zang 2024	1	1	6,546	scRNA:6546	pons:1	unknown:1	0.710
DeSisto 2024	2	2	1,210	scRNA:1210	thalamus:2	unknown:2	0.631
Total	57	56	311,618	18,882 genes shared by all studies			

Malignant-cell calling agrees with published annotations in nine of ten cells

Malignant status was assigned per cell by combining lineage-marker scoring with an expression-based copy-number profile computed against within-sample immune and oligodendrocyte references[19], smoothed over 100-gene windows and summarised both chromosome-wide and per chromosome arm (Methods). We evaluated the calls against the original authors' annotations wherever they exist: **324,676 annotated cells across 97 samples**, with agreement computed per sample so that large cohorts cannot dominate (Table 2). The median per-sample agreement is **0.898 (IQR 0.793–0.978)**; the unweighted mean is 0.836 (95% CI 0.790–0.877) and the cell-weighted pooled value 0.884 (precision 0.934, recall 0.909), the gap reflecting poorer performance in small Smart-seq2 plates and low-complexity nuclei samples (Supplementary Figure 1). Thresholds were tuned on 22 samples; agreement on those samples (0.882) is indistinguishable from the 75 held-out ones (0.885), so the calls are not an artefact of tuning. Calls supported by at least one recurrent arm-level event are more reliable than score-only calls (precision 0.952 versus 0.862), and we therefore ship the event support as a confidence field rather than a filter.

Recurrent events behave as expected for the disease: chromosome 1q gain is the most frequent, followed by 14q and 13q loss, with 17p loss confined to a minority of samples (Figure 1f). We explicitly dropped 4q12 (*PDGFRA*/*KIT*) from the hallmark set: at 100-gene resolution that band is covered by at most one window, and what it measures is OPC-lineage expression against an immune reference rather than copy number — the apparent 4q12 gain frequencies in the TPM-based cohorts were lineage expression, and gene-level deviations are released for exploratory use only. Combining marker and copy-number evidence leaves 182,924 cells (53.5%) called malignant, 60,026 normal and 99,263 (29.0%) ambiguous; a permissive variant that trusts event-backed copy-number calls over discordant markers recovers a further 30,463 cells and is released as a sensitivity label.

Table 2: Agreement of malignant calls with published annotations. Tier-2 comparison of the copy-number call against each study's own annotation, for every annotated cell. Pooled values weight cells equally; the sample median (with interquartile range) and unweighted sample mean weight samples equally.

Study	Cells	Samples	Agreement	Precision	Recall	Sample median (IQR)	Sample mean
Jessa 2022 (sc/snRNA)	84,194	17	0.915	0.982	0.914	0.898 (0.835–0.964)	0.846
Jessa 2022 (multiome)	46,618	10	0.935	0.948	0.967	0.948 (0.897–0.988)	0.930
Collot 2026	17,750	5	0.907	0.896	0.975	0.960 (0.919–0.964)	0.929
Okada 2026	161,043	22	0.850	0.906	0.890	0.868 (0.763–0.895)	0.800
Liu 2022 (query)	8,705	42	0.799	0.975	0.782	0.945 (0.770–0.990)	0.813
LaBelle (query)	6,366	1	1.000	nan	nan	1.000 (1.000–1.000)	1.000
All	**324,676**	**97**	**0.884**	**0.934**	**0.909**	**0.898 (0.793–0.978)**	**0.836**

Lineage labels transfer to held-out patients and to five external cohorts

A scANVI classifier[30] trained on all labelled reference cells, with the validation split made by **patient** rather than by cell, reaches 0.993 training and **0.975 held-out-patient accuracy** (0.917 balanced accuracy) over 23,827 labelled cells from five patients never seen in training. Per-class recall is 0.999 malignant, 0.969 neuron, 0.950 myeloid, 0.917 oligodendrocyte, 0.896 T/NK, 0.883 vascular and 0.803 astrocyte; T/NK precision is the weakest figure at 0.561, reflecting how few lymphocytes these tumours contain (Supplementary Figure 2).

We then projected five cohorts deliberately excluded from training, encoding each with the frozen reference encoder and transferring labels with no fine-tuning (Figure 2a–b). Two are DMG cohorts deposited as transcripts per million rather than counts — Filbin 2018 (3,037 cells, 9 samples)[15] and Liu 2022 (12,438 cells, 42 plates)[18] — and both land predominantly in the malignant territory (median 90.3% and 83.6% of cells per sample); for Liu, cells the authors called malignant are recovered as malignant in 99.3% of cases (Figure 2c). An immune-focused CAR-T cohort projects as roughly half malignant and half immune, as its composition implies. The two normal-brain references provide the specificity test: Siletti 2023 adult pons nuclei[36] are called malignant in a median of only **1.2%** of cells per dissection (1.8% pooled), with adult oligodendrocytes, astrocytes, microglia and neuronal superclusters recovered as their matching compartments, whereas Braun 2023 first-trimester hindbrain[37] projects as 91.3% malignant per sample (87.7%

pooled) (Figure 2d).

That last number is a property of the reference, not an error: the reference contains no normal progenitor class, so fetal radial glia, glioblasts and neuroblasts have nowhere else to go. Their nearest neighbours are informative nonetheless — malignant cells sit closest to fetal glioblast (54.7%) and oligodendroglial (32.4%) classes, a 28.5-fold enrichment for the oligodendroglial class over its frequency in the fetal reference — but the placement is made at a median 15-nearest-neighbour distance of 3.28 in latent space versus 1.57 within the reference, so this is a nearest-neighbour statement in an off-manifold region and not cell-of-origin evidence (Supplementary Figure 3). We release the per-cell neighbour distance and uncertainty percentile so that downstream use can be filtered accordingly.

Only two of fifteen expression metaprograms replicate across studies and assays

We derived expression programmes per sample by consensus non-negative matrix factorisation[38] on 167,012 malignant cells from 50 samples, then clustered the resulting programmes into metaprograms. At the primary dendrogram cut, 15 metaprograms absorb 161 programmes (Table 3, Figure 3a, Supplementary Figure 4). Twelve carry a recognisable lineage identity — four AC-like, two OPC-like, one OC-like, one MES-like, two cycling, one hypoxia and one ciliated programme — and three are technical: ribosomal (MP2), mitochondrial (MP12) and an immediate-early-gene dissociation-stress programme (MP13).

Before deriving them we fixed a replication rule: a metaprogram counts as replicated only if it recurs in at least three studies **and** in both whole cells and nuclei. Under that rule **2 of 15 replicate** — cycling (MP1, 34 samples) and hypoxia (MP4, 15 samples) — and the count is stable across dendrogram cuts from 0.84 to 0.90, rising to 3 only at 0.91 and 5 at 0.92 (Supplementary Figure 4c). The explanation is structural rather than statistical: **every lineage-identity programme in this cohort is nuclei-only**, and the single programme found only in whole cells is the dissociation-stress programme. Masking the 5,921 genes that differ significantly between whole cells and nuclei in the

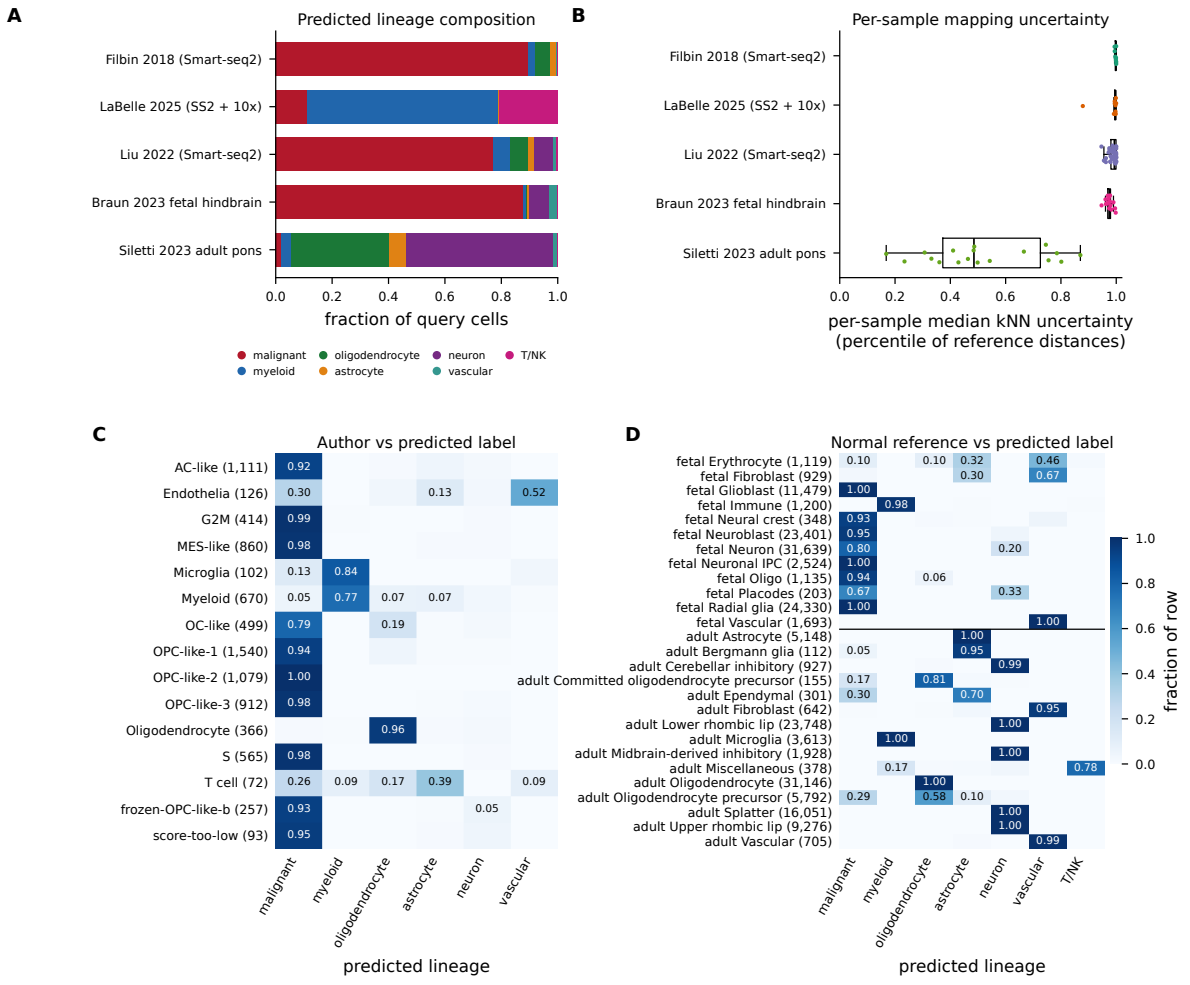


Figure 2: Zero-shot projection of five external cohorts. **a**, Predicted lineage composition of each query cohort. **b**, Per-sample median neighbour-based uncertainty, expressed as a percentile of the reference's own 15-nearest-neighbour distance distribution. **c**, Author-annotated state against predicted lineage for Liu 2022 (row-normalised). **d**, Fetal (Braun 2023) and adult (Siletti 2023) reference classes against predicted lineage; the horizontal rule separates the two references.

two paired multiome samples (plus 92 mitochondrial and ribosomal genes) and repeating the derivation recovers 12 metaprograms with 3 replicated, against a median of 2.5 metaprograms and 1 replicated over ten size-matched random gene masks, so the assay dependence is specific to genes that differ between cells and nuclei rather than a generic effect of removing genes (Figure 3b, Supplementary Figure 5).

Metaprogram families assigned per cell by within-sample standardised scores are distributed similarly in cells and nuclei (within 0.03 per family), which is the correct comparison; the naive between-assay comparison is determined by assay and is not interpretable (Figure 3c–d).

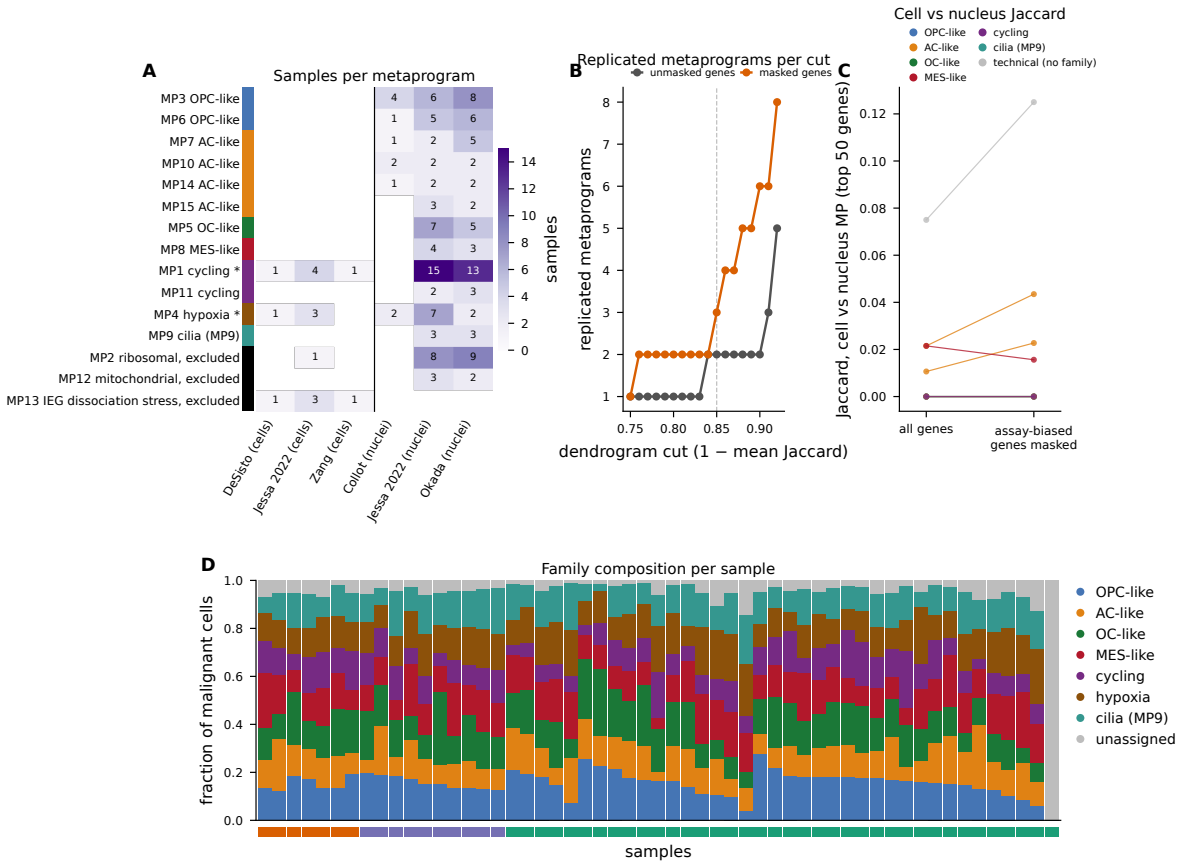


Figure 3: Metaprograms and their assay dependence. **a**, Metaprogram by study-and-assay occurrence matrix; replicated metaprograms are marked and technical programmes are labelled by the class they capture. **b**, Number of replicated metaprograms as a function of dendrogram cut, with and without masking the 6,013 assay-biased genes; the dashed line marks the primary cut. **c**, Jaccard similarity between matched cell-derived and nucleus-derived metaprograms, by family, before and after assay masking. **d**, Per-sample metaprogram-family composition of malignant cells, ordered by study.

Table 3: Consensus metaprograms. Metaprograms at the primary dendrogram cut, their assigned lineage family or technical class, the number of samples in which a contributing programme was found, whether they satisfy the pre-registered replication rule (at least three studies and both assay classes), and their top genes.

Metaprogram	Family	Samples	Replicated	Top genes
MP1	cycling	34	yes	*50*
MP2	*ribosomal*	18	no	*50*
MP3	OPC-like	18	no	*50*
MP4	hypoxia	15	yes	*50*
MP5	OC-like	12	no	*50*
MP6	OPC-like	12	no	*50*
MP7	AC-like	8	no	*50*
MP8	MES-like	7	no	*50*
MP9	ciliated (MP9)	6	no	*50*
MP10	AC-like	6	no	*50*
MP11	cycling	5	no	*50*
MP12	*mitochondrial*	5	no	*41*
MP13	*IEG / dissociation stress*	5	no	*45*
MP14	AC-like	5	no	*50*
MP15	AC-like	5	no	*50*

Microenvironment populations and the confounding of H3 variant with site

Microenvironment subclustering resolves 21 populations — homeostatic microglia, lipid-phagocytic and hypoxic/glycolytic tumour-associated macrophages, interferon-responsive macrophages, proliferating myeloid cells, CD4/CD8/NK lymphocyte states, five endothelial and pericyte populations, and four mature oligodendrocyte states plus a committed OPC population (Figure 4a, Supplementary Figure 6). The myeloid-dominated, lymphocyte-poor composition is consistent with previous descriptions of this microenvironment[21,22]. Because several of these populations are recovered almost entirely from nuclei, we quantified assay purity directly and report it alongside the populations (Figure 4b).

Composition modelling across samples finds one effect that survives adjustment: neurons are more abundant in H3.1 than H3.3 tumours ($q = 0.004$; site-adjusted $q = 0.017$; pons-only $q = 0.050$), whereas astrocyte, myeloid, vascular and malignant-fraction differ-

ences do not survive site adjustment ($q = 0.17$–0.27) or restriction to pons ($q = 0.28$–0.29) — all eight H3.1 samples are pontine nuclei samples, so these contrasts are confounded by design (Figure 4c). The strongest compositional signal in the entire analysis is technical: neurons differ between nuclei and whole cells with a coefficient of 3.40 ($q < 10^{-9}$).

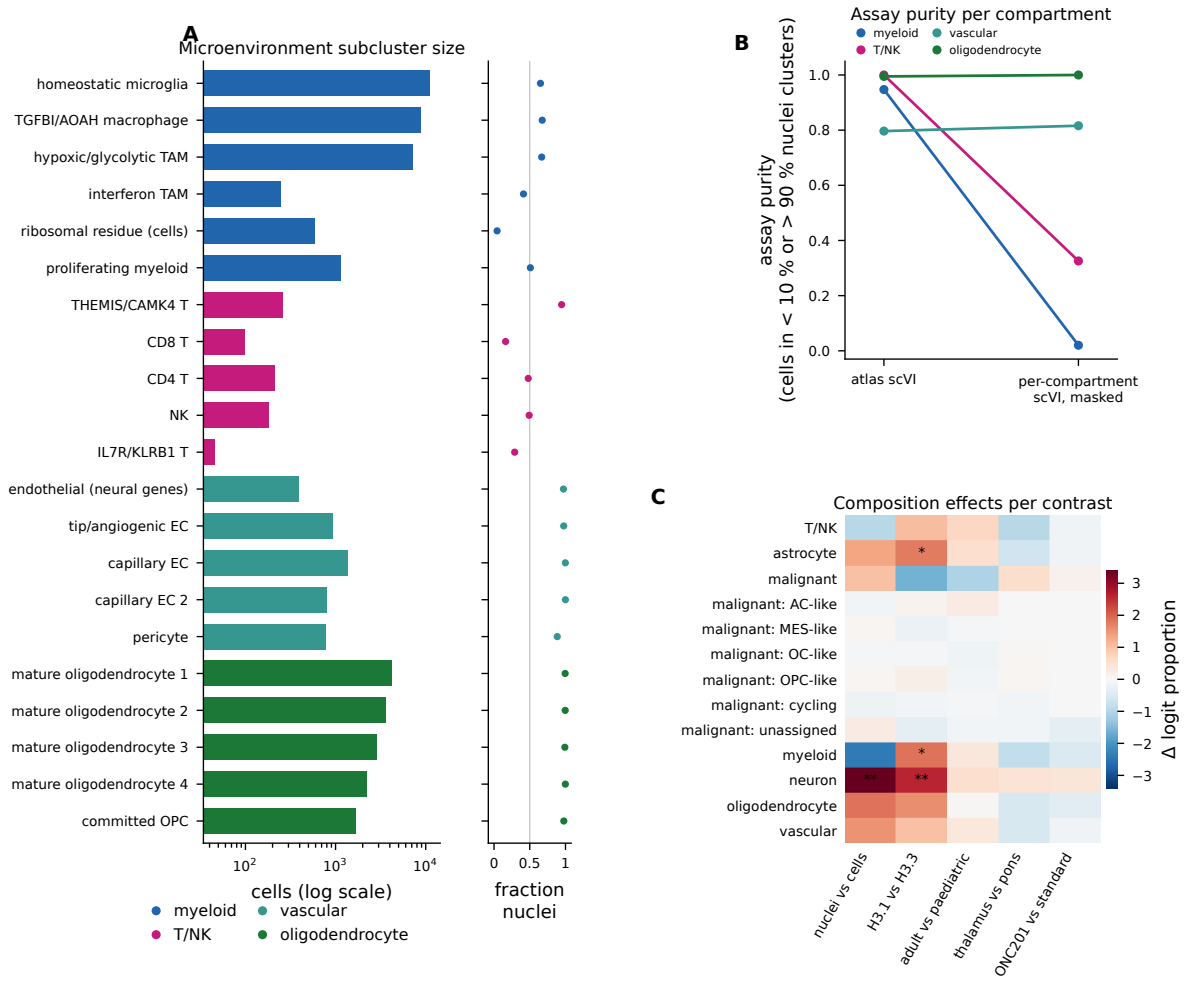


Figure 4: Microenvironment populations and sample-level composition. a, Microenvironment subclusters: cell count (log scale) and the fraction of each cluster derived from nuclei. **b**, Assay purity of microenvironment clusters in the atlas embedding versus a per-compartment model. **c**, Composition contrasts as the change in logit proportion per cell type and malignant state; asterisks mark Benjamini–Hochberg significance[39], and the nuclei-versus-cells column is the technical reference contrast.

Spatial co-localisation of malignant states with normal lineages

We deconvolved 27 Visium sections from four cohorts with cell2location[40] against a single 13-class reference built from the corrected metaprogram families (seven malignant families, six normal lineages), and tested co-localisation on spot **proportions** with **patients** as

the unit of inference: 21 DMG sections from 12 patients, Fisher-z averaged within patient and pooled by random effects, with Benjamini–Hochberg correction across 42 state-by-microenvironment pairs (Table 4, Figure 5).

Astrocyte-like tumour co-localises with normal astrocytes ($\rho = 0.393$, 95% CI 0.234–0.531, $q = 10^{-4}$, positive in 13 of 21 sections) and with oligodendrocytes ($\rho = 0.308$, $q = 0.005$). The ciliated programme behaves similarly, pairing with astrocytes ($\rho = 0.369$, $q = 10^{-4}$) and showing the most consistent positive sign of any pair with neurons ($\rho = 0.230$, $q = 10^{-4}$, positive in 16 of 21 sections). Of our two pre-stated hypotheses, the mesenchymal one holds — MES-like tumour co-localises with myeloid cells ($\rho = 0.238$, CI 0.071–0.392, $q = 0.034$, 14 of 21 sections) and with vasculature ($\rho = 0.253$, $q = 0.034$) — and the oligodendroglial one fails: OPC-like tumour and oligodendrocytes are not associated after correction ($\rho = 0.178$, CI -0.028 to 0.370, $q = 0.210$, 10 of 21 sections). Between-patient heterogeneity is high for every pair ($I^2 = 0.952$–0.997), so these are small, consistent-in-sign effects rather than strong ones, and analysing abundances instead of proportions inflates them by roughly 30% because every class correlates with spot depth (Supplementary Figure 7).

Ten serial sections through one whole-brainstem autopsy allow a within-tumour gradient to be described, though with one patient it remains descriptive: within sections, tumour fraction rises with the astrocyte-like ($\rho = 0.285$, CI 0.149–0.410, positive in 9 of 10 sections) and OPC-like ($\rho = 0.213$, $q = 0.001$) share and falls with the oligodendrocyte (-0.298), astrocyte (-0.431), vascular (-0.235), neuronal (-0.239), mesenchymal (-0.385) and hypoxic (-0.185) shares. Finally, five DMG and five glioblastoma Visium sections deposited in a single study[24] permit a like-for-like comparison: DMG sections carry a higher median tumour fraction (0.528 versus 0.460) and a lower cycling share (0.112 versus 0.214), but with five sections per group neither survives correction ($q = 0.667$ for both), and we report them as not established.

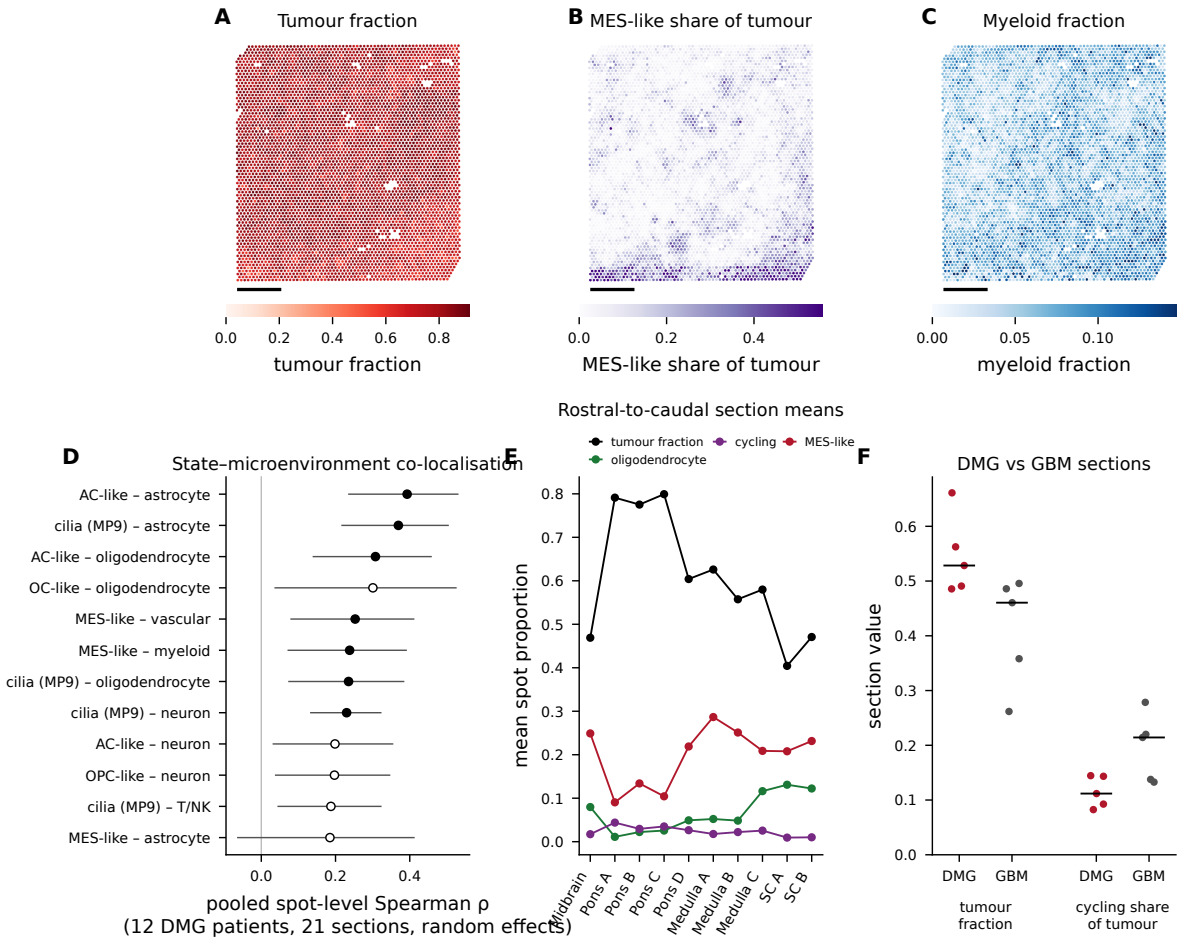


Figure 5: Spatial organisation of malignant states. **a**–**c**, One representative section (pixel coordinates, 1 mm scale bar) showing tumour fraction, the mesenchymal share of tumour and the myeloid fraction. **d**, Pooled spot-level Spearman correlations between malignant states and microenvironment lineages over 12 DMG patients and 21 sections; filled points pass Benjamini–Hochberg correction across 42 pairs. **e**, Mean spot proportions along ten serial sections of one brainstem autopsy. **f**, Tumour fraction and cycling share per section in five DMG and five glioblastoma sections from one study.

Table 4: Spatial co-localisation of malignant states with microenvironment lineages. Spot-level Spearman correlations of cell2location proportions, Fisher-z averaged within patient and pooled by random effects over 12 DMG patients and 21 sections, Benjamini–Hochberg corrected across 42 state-by-microenvironment pairs. The ten strongest pairs are shown; I^2 quantifies between-patient heterogeneity.

Malignant state	Lineage	ρ	95% CI	q	Sections > 0	I^2
AC-like	astrocyte	0.393	0.234 to 0.531	9.38e-05	13/21	0.993
ciliated (MP9)	neuron	0.230	0.131 to 0.324	9.38e-05	16/21	0.978
ciliated (MP9)	astrocyte	0.369	0.216 to 0.505	9.38e-05	13/21	0.992
AC-like	oligodendrocyte	0.308	0.138 to 0.459	0.00512	11/21	0.993
MES-like	vascular	0.253	0.078 to 0.412	0.0339	13/21	0.993
MES-like	myeloid	0.238	0.071 to 0.392	0.0339	14/21	0.992
ciliated (MP9)	oligodendrocyte	0.235	0.073 to 0.385	0.0339	13/21	0.992
OC-like	vascular	-0.214	-0.367 to -0.051	0.0511	6/21	0.993
ciliated (MP9)	T/NK	0.188	0.044 to 0.324	0.0511	14/21	0.990
hypoxia	T/NK	0.082	0.017 to 0.147	0.0589	12/21	0.952

Discussion

This resource makes three contributions. The first is the reference itself: 311,618 cells and nuclei from 52 patients with harmonised labels, calibrated malignant calls, a frozen encoder for projection and a validated held-out-patient classifier. For a tumour where individual cohorts contain a handful of patients, a patient-level denominator of 52 is the difference between describing a sample and describing a disease.

The second is a set of calibrated negative results. Only two of fifteen metaprograms replicate under a rule fixed in advance, and the reason is not noise but confounding: lineage-identity programmes appear in nuclei and the dissociation-stress programme appears in cells, so studies profiling different material will recover different states and agree only on cycling and hypoxia. This is a sharper version of the warning that emerges from pan-cancer programme analysis[28]: recurrence across tumours is necessary but not sufficient when the cohorts differ systematically in how the material was prepared. One of our two pre-stated spatial hypotheses survived patient-level correction and the other did not. A compositional difference between H3.1 and H3.3 tumours is present for neurons and disappears for every other cell type once site is adjusted, because H3.1 sampling is pontine sampling in these deposits. We regard these as the most immediately useful outputs for anyone planning a DMG single-cell study: they identify which comparisons

are currently decidable from public data and which require matched material.

The third is methodological. Agreement between an expression-based copy-number call and the original authors' annotations is not uniform — a median of 0.898 per sample with an interquartile range spanning nearly twenty percentage points — and reporting it per sample rather than pooled is what exposes that. Event-backed and score-only calls differ by nine points of precision and should not be treated alike. Zero-shot projection into a frozen reference is accurate for tumour and adult-brain queries, and its failure mode on fetal tissue is interpretable and detectable from the neighbour distances we release.

Several biological observations deserve follow-up in matched material. The ciliated programme is the clearest candidate: it is the most consistently positive partner of normal neurons across sections and pairs with astrocytes as strongly as the astrocyte-like programme does, yet it appears in no established DMG state nomenclature[15,16,18]. The co-localisation of mesenchymal tumour with myeloid cells and vasculature is consistent with the hypoxia-associated mesenchymal compartment described in glioblastoma[20] and with BMP-driven invasive phenotypes reported in DMG[25], and the brainstem gradient — tumour-rich regions enriched for astrocyte-like and OPC-like programmes and depleted of normal glia — is what an infiltrative front is expected to look like, but a single autopsy cannot establish it. Engineered human models that generate H3.3K27M-altered DMG *de novo* in brainstem organoids[41] now offer a route to test whether these associations are causal rather than observational.

Limitations of the study

One autopsy series contributes half the reference cells, and every autopsy sample in the atlas comes from it, so autopsy versus resection is not an estimable contrast. All eight H3.1 samples are pontine nuclei samples, and the H3 variant is unreported for 25 of 57 samples. The whole-cell arm of any assay comparison rests on six samples. Copy-number thresholds were tuned on 22 samples, and 18.9% of malignant calls rest on the aggregate score with no supporting arm-level event. The reference contains no normal progenitor population, so developmental queries project into the malignant region by

construction. The spatial conclusions come from 21 sections and 12 patients with high between-patient heterogeneity, the brainstem gradient from one patient, and the DMG-versus-glioblastoma comparison from five sections per group. Two of the eleven cohorts are deposited as transcripts per million with no recoverable counts and are used for projection only. Metaprogram derivation was limited to nine programmes per sample in 21 of 50 samples by cell number, so finer programmes could not be resolved.

Resource availability

Lead contact

Requests for further information should be directed to the lead contact, [name] ([email]).

Materials availability

This study did not generate new reagents or materials.

Data and code availability

- This study reanalyses published human data only. All primary accessions are listed in the key resources table (Table 5) and Supplementary Table 1.
- The integrated object (`dmg_atlas_v1.h5ad`, CELLxGENE schema 7.x, validated with zero MUST failures) carries raw counts, harmonized metadata, lineage and state labels, malignant calls with confidence fields, and all model embeddings. *Release of the merged count matrix is pending confirmation of redistribution terms for the post-mortem cohort, whose raw data is under controlled access (EGAS50000001447); if those terms preclude redistribution, the release will ship embeddings, labels and metadata with per-cohort pointers to the primary deposits.*
- All analysis code, the phase-by-phase documentation and the complete audit trail are available in the project repository [URL]; every figure panel records the md5 checksum of each input table.

- Any additional information required to reanalyse the data reported in this paper is available from the lead contact on request.

Acknowledgments

We thank the investigators and patients of every cohort reused here; this work exists only because those data were deposited publicly.

References

[1] David N Louis, Arie Perry, Pieter Wesseling, Daniel J Brat, Ian A Cree, Dominique Figarella-Branger, Cynthia Hawkins, H K Ng, Stefan M Pfister, Guido Reifenberger, Riccardo Soffietti, Andreas von Deimling, and David W Ellison. The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. *Neuro-oncology*, 23(8):1231–1251, 2021. doi: 10.1093/neuonc/noab106.

[2] Alan Mackay, Anna Burford, Diana Carvalho, Elisa Izquierdo, Janat Fazal-Salom, Kathryn R Taylor, Lynn Bjerke, Matthew Clarke, Mara Vinci, Meera Nandhabalan, Sara Temelso, Sergey Popov, Valeria Molinari, Pichai Raman, Angela J Waanders, Harry J Han, Saumya Gupta, Lynley Marshall, Stergios Zacharoulis, Sucheta Vaidya, Henry C Mandeville, Leslie R Bridges, Andrew J Martin, Safa Al-Sarraj, Christopher Chandler, Ho-Keung Ng, Xingang Li, Kun Mu, Saoussen Trabelsi, Dorra H'mida-Ben Brahim, Alexei N Kisljakov, Dmitry M Konovalov, Andrew S Moore, Angel Montero Carcaboso, Mariona Sunol, Carmen de Torres, Ofelia Cruz, Jaume Mora, Ludmila I Shats, João N Stavale, Lucas T Bidinotto, Rui M Reis, Natacha Entz-Werle, Michael Farrell, Jane Cryan, Darach Crimmins, John Caird, Jane Pears, Michelle Monje, Marie-Anne Debily, David Castel, Jacques Grill, Cynthia Hawkins, Hamid Nikbakht, Nada Jabado, Suzanne J Baker, Stefan M Pfister, David T W Jones, Maryam Fouladi, André O von Bueren, Michael Baudis, Adam Resnick, and Chris Jones. Integrated Molecular Meta-Analysis of 1,000 Pediatric High-Grade and Diffuse Intrinsic Pontine Glioma. *Cancer cell*, 32(4):520–537.e5, 2017. doi: 10.1016/j.ccell.2017.08.017.

[3] Lindsey M Hoffman, Sophie E M Veldhuijzen van Zanten, Niclas Colditz, Joshua Baugh, Brooklyn Chaney, Marion Hoffmann, Adam Lane, Christine Fuller, Lili Miles, Cynthia Hawkins, Ute Bartels, Eric Bouffet, Stewart Goldman, Sarah Leary, Nicholas K Foreman, Roger Packer, Katherine E Warren, Alberto Broniscer, Mark W Kieran, Jane Minturn, Melanie Comito, Emmett Broxson, Chie-Schin Shih, Soumen Khatua, Murali Chintagumpala, Anne Sophie Carret, Nancy Yanez Escorza, Timothy Hassall, David S Ziegler, Nicholas Gottardo, Hetal Dholaria, Renee Doughman, Martin Benesch, Rachid Drissi, Javad Nazarian, Nada Jabado, Nathalie Boddaert, Pascale Varlet, Géraldine Giraud, David Castel, Stephanie Puget, Chris Jones, Esther Hulleman, Piergiorgio Modena, Marzia Giagnacovo, Manila Antonelli, Torsten Pietsch, Gerrit H Gielen, David T W Jones, Dominik Sturm, Stefan M Pfister, Nicolas U Gerber, Michael A Grotzer, Elke Pfaff, André O von Bueren, Darren Hargrave, Guirish A Solanki, Filip Jadrijevic Cvrlje, Gertjan J L Kaspers, William P Vandertop, Jacques Grill, Simon Bailey, Veronica Biassoni, Maura Massimino, Raphaël Calmon, Esther Sanchez, Brigitte Bison, Monika Warmuth-Metz, James Leach, Blaise Jones, Dannis G van Vuurden, Christof M Kramm, and Maryam Fouladi. Clinical, Radiologic, Pathologic, and Molecular Characteristics of Long-Term Survivors of Diffuse Intrinsic Pontine Glioma (DIPG): A Collaborative Report From the International and European Society for Pediatric Oncology DIPG Registries. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*, 36 (19):1963–1972, 2018. doi: 10.1200/JCO.2017.75.9308.

[4] Jeremy Schwartzentruber, Andrey Korshunov, Xiao-Yang Liu, David T W Jones, Elke Pfaff, Karine Jacob, Dominik Sturm, Adam M Fontebasso, Dong-Anh Khuong Quang, Martje Tönjes, Volker Hovestadt, Steffen Albrecht, Marcel Kool, Andre Nantel, Carolin Konermann, Anders Lindroth, Natalie Jäger, Tobias Rausch, Marina Ryzhova, Jan O Korbel, Thomas Hielscher, Peter Hauser, Miklos Garami, Almos Klekner, Laszlo Bognar, Martin Ebinger, Martin U Schuhmann, Wolfram Scheurlen, Arnulf Pekrun, Michael C Frühwald, Wolfgang Roggendorf, Christoph Kramm, Matthias Dürken, Jeffrey Atkinson, Pierre Lepage, Alexandre Montpetit,

Magdalena Zakrzewska, Krzystof Zakrzewski, Pawel P Liberski, Zhifeng Dong, Peter Siegel, Andreas E Kulozik, Marc Zapatka, Abhijit Guha, David Malkin, Jörg Felsberg, Guido Reifenberger, Andreas von Deimling, Koichi Ichimura, V Peter Collins, Hendrik Witt, Till Milde, Olaf Witt, Cindy Zhang, Pedro Castelo-Branco, Peter Lichter, Damien Faury, Uri Tabori, Christoph Plass, Jacek Majewski, Stefan M Pfister, and Nada Jabado. Driver mutations in histone H3.3 and chromatin remodelling genes in paediatric glioblastoma. *Nature*, 482(7384):226–31, 2012. doi: 10.1038/nature10833.

[5] Gang Wu, Alberto Broniscer, Troy A McEachron, Charles Lu, Barbara S Paugh, Jared Becksfort, Chunxu Qu, Li Ding, Robert Huether, Matthew Parker, Junyuan Zhang, Amar Gajjar, Michael A Dyer, Charles G Mullighan, Richard J Gilbertson, Elaine R Mardis, Richard K Wilson, James R Downing, David W Ellison, Jinghui Zhang, and Suzanne J Baker. Somatic histone H3 alterations in pediatric diffuse intrinsic pontine gliomas and non-brainstem glioblastomas. *Nature genetics*, 44(3): 251–3, 2012. doi: 10.1038/ng.1102.

[6] Peter W Lewis, Manuel M Müller, Matthew S Koletsky, Francisco Cordero, Shu Lin, Laura A Banaszynski, Benjamin A Garcia, Tom W Muir, Oren J Becher, and C David Allis. Inhibition of PRC2 activity by a gain-of-function H3 mutation found in pediatric glioblastoma. *Science (New York, N.Y.)*, 340(6134):857–61, 2013. doi: 10.1126/science.1232245.

[7] Sebastian Bender, Yujie Tang, Anders M Lindroth, Volker Hovestadt, David T W Jones, Marcel Kool, Marc Zapatka, Paul A Northcott, Dominik Sturm, Wei Wang, Bernhard Radlwimmer, Jonas W Højfeldt, Nathalène Truffaux, David Castel, Simone Schubert, Marina Ryzhova, Huriye Seker-Cin, Jan Gronych, Pascal David Johann, Sebastian Stark, Jochen Meyer, Till Milde, Martin Schuhmann, Martin Ebinger, Camelia-Maria Monoranu, Anitha Ponnuswami, Spenser Chen, Chris Jones, Olaf Witt, V Peter Collins, Andreas von Deimling, Nada Jabado, Stephanie Puget, Jacques Grill, Kristian Helin, Andrey Korshunov, Peter Lichter, Michelle Monje, Christoph Plass, Yoon-Jae Cho, and Stefan M Pfister. Reduced H3K27me3 and DNA

hypomethylation are major drivers of gene expression in K27M mutant pediatric high-grade gliomas. *Cancer cell*, 24(5):660–72, 2013. doi: 10.1016/j.ccr.2013.10.006.

[8] Ashot S Harutyunyan, Brian Krug, Haifen Chen, Simon Papillon-Cavanagh, Michele Zeinieh, Nicolas De Jay, Shriya Deshmukh, Carol C L Chen, Jad Belle, Leonie G Mikael, Dylan M Marchione, Rui Li, Hamid Nikbakht, Bo Hu, Gael Cagnone, Warren A Cheung, Abdulshakour Mohammadnia, Denise Bechet, Damien Faury, Melissa K McConechy, Manav Pathania, Siddhant U Jain, Benjamin Ellezam, Alexander G Weil, Alexandre Montpetit, Paolo Salomoni, Tomi Pastinen, Chao Lu, Peter W Lewis, Benjamin A Garcia, Claudia L Kleinman, Nada Jabado, and Jacek Majewski. H3K27M induces defective chromatin spread of PRC2-mediated repressive H3K27me2/me3 and is essential for glioma tumorigenesis. *Nature communications*, 10(1):1262, 2019. doi: 10.1038/s41467-019-09140-x.

[9] Brian Krug, Nicolas De Jay, Ashot S Harutyunyan, Shriya Deshmukh, Dylan M Marchione, Paul Guilhamon, Kelsey C Bertrand, Leonie G Mikael, Melissa K McConechy, Carol C L Chen, Sima Khazaei, Robert F Koncar, Sameer Agnihotri, Damien Faury, Benjamin Ellezam, Alexander G Weil, Josie Ursini-Siegel, Daniel D De Carvalho, Peter B Dirks, Peter W Lewis, Paolo Salomoni, Mathieu Lupien, Cheryl Arrowsmith, Paul F Lasko, Benjamin A Garcia, Claudia L Kleinman, Nada Jabado, and Stephen C Mack. Pervasive H3K27 Acetylation Leads to ERV Expression and a Therapeutic Vulnerability in H3K27M Gliomas. *Cancer cell*, 35(5):782–797.e8, 2019. doi: 10.1016/j.ccell.2019.04.004.

[10] Surya Nagaraja, Nicholas A Vitanza, Pamelyn J Woo, Kathryn R Taylor, Fang Liu, Lei Zhang, Meng Li, Wei Meng, Anitha Ponnuswami, Wenchao Sun, Jie Ma, Esther Hulleman, Tomek Swigut, Joanna Wysocka, Yujie Tang, and Michelle Monje. Transcriptional Dependencies in Diffuse Intrinsic Pontine Glioma. *Cancer cell*, 31 (5):635–652.e6, 2017. doi: 10.1016/j.ccell.2017.03.011.

[11] Humsa S Venkatesh, Wade Morishita, Anna C Geraghty, Dana Silverbush, Shawn M Gillespie, Marlene Arzt, Lydia T Tam, Cedric Espenel, Anitha Ponnuswami, Lijun Ni, Pamelyn J Woo, Kathryn R Taylor, Amit Agarwal, Aviv Regev, David Brang,

Hannes Vogel, Shawn Hervey-Jumper, Dwight E Bergles, Mario L Suvà, Robert C Malenka, and Michelle Monje. Electrical and synaptic integration of glioma into neural circuits. *Nature*, 573(7775):539–545, 2019. doi: 10.1038/s41586-019-1563-y.

[12] Mara Vinci, Anna Burford, Valeria Molinari, Ketty Kessler, Sergey Popov, Matthew Clarke, Kathryn R Taylor, Helen N Pemberton, Christopher J Lord, Alice Gutteridge, Tim Forshew, Diana Carvalho, Lynley V Marshall, Elizabeth Y Qin, Wendy J Ingram, Andrew S Moore, Ho-Keung Ng, Saoussen Trabelsi, Dorra H'mida-Ben Brahim, Natacha Entz-Werle, Stergios Zacharoulis, Sucheta Vaidya, Henry C Mandeville, Leslie R Bridges, Andrew J Martin, Safa Al-Sarraj, Christopher Chandler, Mariona Sunol, Jaume Mora, Carmen de Torres, Ofelia Cruz, Angel M Carcaboso, Michelle Monje, Alan Mackay, and Chris Jones. Functional diversity and cooperativity between subclonal populations of pediatric glioblastoma and diffuse intrinsic pontine glioma cells. *Nature medicine*, 24(8):1204–1215, 2018. doi: 10.1038/s41591-018-0086-7.

[13] Isabel Arrillaga-Romany, Sharon L Gardner, Yazmin Odia, Dolly Aguilera, Joshua E Allen, Tracy Batchelor, Nicholas Butowski, Clark Chen, Timothy Cloughesy, Andrew Cluster, John de Groot, Karan S Dixit, Jerome J Graber, Aya M Haggiagi, Rebecca A Harrison, Albert Kheradpour, Lindsay B Kilburn, Sylvia C Kurz, Guangrong Lu, Tobey J MacDonald, Minesh Mehta, Allen S Melemed, Phioanh Leia Nghiemphu, Samuel C Ramage, Nicole Shonka, Ashley Sumrall, Rohinton S Tarapore, Lynne Taylor, Yoshie Umemura, and Patrick Y Wen. ONC201 (Dordaviprone) in Recurrent H3 K27M-Mutant Diffuse Midline Glioma. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*, 42(13):1542–1552, 2024. doi: 10.1200/JCO.23.01134.

[14] Robbie G Majzner, Sneha Ramakrishna, Kristen W Yeom, Shabnum Patel, Harshini Chinnasamy, Liora M Schultz, Rebecca M Richards, Li Jiang, Valentin Barsan, Rebecca Mancusi, Anna C Geraghty, Zinaida Good, Aaron Y Mochizuki, Shawn M Gillespie, Angus Martin Shaw Toland, Jasia Mahdi, Agnes Reschke, Esther H Nie, Isabelle J Chau, Maria Caterina Rotiroti, Christopher W Mount, Christina Bag-

gott, Sharon Mavroukakis, Emily Egeler, Jennifer Moon, Courtney Erickson, Sean Green, Michael Kunicki, Michelle Fujimoto, Zach Ehlinger, Warren Reynolds, Sreevidya Kurra, Katherine E Warren, Snehit Prabhu, Hannes Vogel, Lindsey Rasmussen, Timothy T Cornell, Sonia Partap, Paul G Fisher, Cynthia J Campen, Mariella G Filbin, Gerald Grant, Bita Sahaf, Kara L Davis, Steven A Feldman, Crystal L Mackall, and Michelle Monje. GD2-CAR T cell therapy for H3K27M-mutated diffuse midline gliomas. *Nature*, 603(7903):934–941, 2022. doi: 10.1038/s41586-022-04489-4.

[15] Mariella G Filbin, Itay Tirosh, Volker Hovestadt, McKenzie L Shaw, Leah E Escalante, Nathan D Mathewson, Cyril Neftel, Nelli Frank, Kristine Pelton, Christine M Hebert, Christine Haberler, Keren Yizhak, Johannes Gojo, Kristof Egervari, Christopher Mount, Peter van Galen, Dennis M Bonal, Quang-De Nguyen, Alexander Beck, Claire Sinai, Thomas Czech, Christian Dorfer, Liliana Goumnerova, Cinzia Lavarino, Angel M Carcaboso, Jaume Mora, Ravindra Mylvaganam, Christina C Luo, Andreas Peyrl, Mara Popović, Amedeo Azizi, Tracy T Batchelor, Matthew P Frosch, Maria Martinez-Lage, Mark W Kieran, Pratiti Bandopadhayay, Rameen Beroukhim, Gerhard Fritsch, Gad Getz, Orit Rozenblatt-Rosen, Kai W Wucherpfennig, David N Louis, Michelle Monje, Irene Slavc, Keith L Ligon, Todd R Golub, Aviv Regev, Bradley E Bernstein, and Mario L Suvà. Developmental and oncogenic programs in H3K27M gliomas dissected by single-cell RNA-seq. *Science (New York, N.Y.)*, 360(6386):331–335, 2018. doi: 10.1126/science.aao4750.

[16] Selin Jessa, Abdulshakour Mohammadnia, Ashot S Harutyunyan, Maud Hulswit, Srinidhi Varadharajan, Hussein Lakkis, Nisha Kabir, Zahedeh Bashardanesh, Steven Hébert, Damien Faury, Maria C Vladoiu, Samantha Worme, Marie Coutelier, Brian Krug, Augusto Faria Andrade, Manav Pathania, Andrea Bajic, Alexander G Weil, Benjamin Ellezam, Jeffrey Atkinson, Roy W R Dudley, Jean-Pierre Farmer, Sebastien Perreault, Benjamin A Garcia, Valérie Larouche, Mathieu Blanchette, Livia Garzia, Aparna Bhaduri, Keith L Ligon, Pratiti Bandopadhayay, Michael D Taylor, Stephen C Mack, Nada Jabado, and Claudia L Kleinman. K27M in canonical and noncanonical H3 variants occurs in distinct oligodendroglial cell lin-

eages in brain midline gliomas. *Nature genetics*, 54(12):1865–1880, 2022. doi: 10.1038/s41588-022-01205-w.

[17] Selin Jessa, Alexis Blanchet-Cohen, Brian Krug, Maria Vladoiu, Marie Coutelier, Damien Faury, Brice Poreau, Nicolas De Jay, Steven Hébert, Jean Monlong, W Todd Farmer, Laura K Donovan, Yixing Hu, Melissa K McConechy, Florence M G Cavalli, Leonie G Mikael, Benjamin Ellezam, Maxime Richer, Andréa Allaire, Alexander G Weil, Jeffrey Atkinson, Jean-Pierre Farmer, Roy W R Dudley, Valerie Larouche, Louis Crevier, Steffen Albrecht, Mariella G Filbin, Hervé Sartelet, Pierre-Eric Lutz, Corina Nagy, Gustavo Turecki, Santiago Costantino, Peter B Dirks, Keith K Murai, Guillaume Bourque, Jiannis Ragoussis, Livia Garzia, Michael D Taylor, Nada Jabado, and Claudia L Kleinman. Stalled developmental programs at the root of pediatric brain tumors. *Nature genetics*, 51(12):1702–1713, 2019. doi: 10.1038/s41588-019-0531-7.

[18] Ilon Liu, Li Jiang, Erik R Samuelsson, Sergio Marco Salas, Alexander Beck, Olivia A Hack, Daeun Jeong, McKenzie L Shaw, Bernhard Englinger, Jenna LaBelle, Hafsa M Mire, Sibylle Madlener, Lisa Mayr, Michael A Quezada, Maria Trissal, Eshini Panditharatna, Kati J Ernst, Jayne Vogelzang, Taylor A Gatesman, Matthew E Halbert, Hana Palova, Petra Pokorna, Jaroslav Sterba, Ondrej Slaby, Rene Geyeregger, Aaron Diaz, Izac J Findlay, Matthew D Dun, Adam Resnick, Mario L Suvà, David T W Jones, Sameer Agnihotri, Jessica Svedlund, Carl Koschmann, Christine Haberler, Thomas Czech, Irene Slavc, Jennifer A Cotter, Keith L Ligon, Sanda Alexandrescu, W K Alfred Yung, Isabel Arrillaga-Romany, Johannes Gojo, Michelle Monje, Mats Nilsson, and Mariella G Filbin. The landscape of tumor cell states and spatial organization in H3-K27M mutant diffuse midline glioma across age and location. *Nature genetics*, 54(12):1881–1894, 2022. doi: 10.1038/s41588-022-01236-3.

[19] Anoop P Patel, Itay Tirosh, John J Trombetta, Alex K Shalek, Shawn M Gillespie, Hiroaki Wakimoto, Daniel P Cahill, Brian V Nahed, William T Curry, Robert L Martuza, David N Louis, Orit Rozenblatt-Rosen, Mario L Suvà, Aviv Regev, and Bradley E Bernstein. Single-cell RNA-seq highlights intratumoral heterogeneity in

primary glioblastoma. *Science (New York, N.Y.)*, 344(6190):1396–401, 2014. doi: 10.1126/science.1254257.

[20] Cyril Neftel, Julie Laffy, Mariella G Filbin, Toshiro Hara, Marni E Shore, Gilbert J Rahme, Alyssa R Richman, Dana Silverbush, McKenzie L Shaw, Christine M Hebert, John Dewitt, Simon Gritsch, Elizabeth M Perez, L Nicolas Gonzalez Castro, Xiaoyang Lan, Nicholas Druck, Christopher Rodman, Danielle Dionne, Alexander Kaplan, Mia S Bertalan, Julia Small, Kristine Pelton, Sarah Becker, Dennis Bonal, Quang-De Nguyen, Rachel L Servis, Jeremy M Fung, Ravindra Mylvaganam, Lisa Mayr, Johannes Gojo, Christine Haberler, Rene Geyeregger, Thomas Czech, Irene Slavc, Brian V Nahed, William T Curry, Bob S Carter, Hiroaki Wakimoto, Priscilla K Brastianos, Tracy T Batchelor, Anat Stemmer-Rachamimov, Maria Martinez-Lage, Matthew P Frosch, Ivan Stamenkovic, Nicolo Riggi, Esther Rheinbay, Michelle Monje, Orit Rozenblatt-Rosen, Daniel P Cahill, Anoop P Patel, Tony Hunter, Inder M Verma, Keith L Ligon, David N Louis, Aviv Regev, Bradley E Bernstein, Itay Tirosh, and Mario L Suvà. An Integrative Model of Cellular States, Plasticity, and Genetics for Glioblastoma. *Cell*, 178(4):835–849.e21, 2019. doi: 10.1016/j.cell.2019.06.024.

[21] Grant L Lin, Surya Nagaraja, Mariella G Filbin, Mario L Suvà, Hannes Vogel, and Michelle Monje. Non-inflammatory tumor microenvironment of diffuse intrinsic pontine glioma. *Acta neuropathologica communications*, 6(1):51, 2018. doi: 10.1186/s40478-018-0553-x.

[22] James L Ross, Montserrat Puigdelloses-Vallcorba, Gonzalo Piñero, Nishant Soni, Wes Thomason, John DeSisto, Angelo Angione, Nadejda M Tsankova, Maria G Castro, Matthew Schniederjan, Nitin R Wadhwani, G Praveen Raju, Peter Morgenstern, Oren J Becher, Adam L Green, Alexander M Tsankov, and Dolores Hambardzumyan. Microglia and monocyte-derived macrophages drive progression of pediatric high-grade gliomas and are transcriptionally shaped by histone mutations. *Immunity*, 57 (11):2669–2687.e6, 2024. doi: 10.1016/j.immuni.2024.09.007.

[23] Raphaël Collot, Cristian Ruiz-Moreno, Celina Honhoff, Thijs J M van den Broek,

Amber K L Wezenaar, Daan J Kloosterman, Hendrikus C R Ariese, Hannah Johnson, Britt M T Vervoort, Amal Jeiroshi, Jens Bunt, Ravian L van Ineveld, Emma Bokobza, Heggert G Rebel, Brigit M Te Pas, Femke C A Ringnalda, Marc van de Wetering, Pierre A Robe, Marcel Kool, Jennifer R Cochran, Mariëtte E G Kranendonk, Dannis G van Vuurden, Esther Hulleman, Ellen J Wehrens, Anoek Zomer, Hendrik G Stunnenberg, and Anne C Rios. IGSF11-VISTA is a critical and targetable immune checkpoint axis in diffuse midline glioma. *Cancer cell*, 44(3):641–657.e9, 2026. doi: 10.1016/j.ccell.2025.12.020.

[24] Yanming Ren, Zongyao Huang, Lingling Zhou, Peng Xiao, Junwei Song, Ping He, Chuanxing Xie, Ran Zhou, Menghan Li, Xiangqun Dong, Qing Mao, Chao You, Jianguo Xu, Yanhui Liu, Zhigang Lan, Tiejun Zhang, Qi Gan, Yuan Yang, Tengyun Chen, Bowen Huang, Xiang Yang, Anqi Xiao, Yun Ou, Zhengzheng Su, Lu Chen, Yan Zhang, Yan Ju, Yuekang Zhang, and Yuan Wang. Spatial transcriptomics reveals niche-specific enrichment and vulnerabilities of radial glial stem-like cells in malignant gliomas. *Nature communications*, 14(1):1028, 2023. doi: 10.1038/s41467-023-36707-6.

[25] Paul Huchede, Swann Meyer, Clément Berthelot, Maud Hamadou, Adrien Bertrand-Chapel, Andria Rakotomalala, Line Manceau, Julia Tomine, Nicolas Lespinasse, Paul Lewandowski, Martine Cordier-Bussat, Laura Broutier, Aurélie Dutour, Isabelle Rochet, Jean-Yves Blay, Cyril Degletagne, Valéry Attignon, Angel Montero-Carcaboso, Marion Le Grand, Eddy Pasquier, Alexandre Vasiljevic, Pascale Gilardi-Hebenstreit, Samuel Meignan, Pierre Leblond, Vanessa Ribes, Erika Cosset, and Marie Castets. BMP2 and BMP7 cooperate with H3.3K27M to promote quiescence and invasiveness in pediatric diffuse midline gliomas. *eLife*, 12, 2024. doi: 10.7554/eLife.91313.

[26] Masahiro Okada, Bavani Subramaniam, Baobao Geng, Jangham Jung, Bohyeon Yu, Felicia Lin, Yu Fu, Yishu Wang, Weiqiang Yu, Sandra Laternser, Nalin Gupta, Sabine Müller, Lin Wang, Aaron Diaz, and Javad Nazarian. Single-nucleus profiling of postmortem diffuse midline gliomas identifies mitochondrial biogenesis as a re-

sistance mechanism to imipridone therapy. *Neuro-oncology*, 28(9):2248–2262, 2026. doi: 10.1093/neuonc/noag119.

[27] John DeSisto, Andrew M Donson, Andrea M Griesinger, Rui Fu, Kent Riemondy, Jean Mulcahy Levy, Julie A Siegenthaler, Nicholas K Foreman, Rajeev Vibhakar, and Adam L Green. Tumor and immune cell types interact to produce heterogeneous phenotypes of pediatric high-grade glioma. *Neuro-oncology*, 26(3):538–552, 2024. doi: 10.1093/neuonc/noad207.

[28] Avishai Gavish, Michael Tyler, Alissa C Greenwald, Rouven Hoefflin, Dor Simkin, Roi Tschernichovsky, Noam Galili Darnell, Einav Somech, Chaya Barbolin, Tomer Antman, Daniel Kovarsky, Thomas Barrett, L Nicolas Gonzalez Castro, Debdatta Halder, Rony Chanoch-Myers, Julie Laffy, Michael Mints, Adi Wider, Rotem Tal, Avishay Spitzer, Toshiro Hara, Maria Raitses-Gurevich, Chani Stossel, Talia Golan, Amit Tirosh, Mario L Suvà, Sidharth V Puram, and Itay Tirosh. Hallmarks of transcriptional intratumour heterogeneity across a thousand tumours. *Nature*, 618 (7965):598–606, 2023. doi: 10.1038/s41586-023-06130-4.

[29] Romain Lopez, Jeffrey Regier, Michael B Cole, Michael I Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature methods*, 15(12): 1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.

[30] Chenling Xu, Romain Lopez, Edouard Mehlman, Jeffrey Regier, Michael I Jordan, and Nir Yosef. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Molecular systems biology*, 17(1):e9620, 2021. doi: 10.15252/msb.20209620.

[31] Mohammad Lotfollahi, Mohsen Naghipourfar, Malte D Luecken, Matin Khajavi, Maren Büttner, Marco Wagenstetter, Žiga Avsec, Adam Gayoso, Nir Yosef, Marta Interlandi, Sergei Rybakov, Alexander V Misharin, and Fabian J Theis. Mapping single-cell data to reference atlases by transfer learning. *Nature biotechnology*, 40(1): 121–130, 2021. doi: 10.1038/s41587-021-01001-7.

[32] Malte D Luecken, M Büttner, K Chaichoompu, A Danese, M Interlandi, M F Mueller, D C Strobl, L Zappia, M Dugas, M Colomé-Tatché, and Fabian J Theis. Benchmark-

ing atlas-level data integration in single-cell genomics. *Nature methods*, 19(1):41–50, 2021. doi: 10.1038/s41592-021-01336-8.

[33] Dongdong Zang, Zilong Dong, Yuecheng Liu, and Qian Chen. Single-cell RNA sequencing of anaplastic ependymoma and H3K27M-mutant diffuse midline glioma. *BMC neurology*, 24(1):74, 2024. doi: 10.1186/s12883-024-03558-7.

[34] Tim Stuart, Andrew Butler, Paul Hoffman, Christoph Hafemeister, Efthymia Papalexi, William M Mauck, Yuhan Hao, Marlon Stoeckius, Peter Smibert, and Rahul Satija. Comprehensive Integration of Single-Cell Data. *Cell*, 177(7):1888–1902.e21, 2019. doi: 10.1016/j.cell.2019.05.031.

[35] Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-Ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0.

[36] Kimberly Siletti, Rebecca Hodge, Alejandro Mossi Albiach, Ka Wai Lee, Song-Lin Ding, Lijuan Hu, Peter Lönnerberg, Trygve Bakken, Tamara Casper, Michael Clark, Nick Dee, Jessica Gloe, Daniel Hirschstein, Nadiya V Shapovalova, C Dirk Keene, Julie Nyhus, Herman Tung, Anna Marie Yanny, Ernest Arenas, Ed S Lein, and Sten Linnarsson. Transcriptomic diversity of cell types across the adult human brain. *Science (New York, N.Y.)*, 382(6667):eadd7046, 2023. doi: 10.1126/science.add7046.

[37] Emelie Braun, Miri Danan-Gotthold, Lars E Borm, Ka Wai Lee, Elin Vinsland, Peter Lönnerberg, Lijuan Hu, Xiaofei Li, Xiaoling He, Žaneta Andrusivová, Joakim Lundeberg, Roger A Barker, Ernest Arenas, Erik Sundström, and Sten Linnarsson. Comprehensive cell atlas of the first-trimester developing human brain. *Science (New York, N.Y.)*, 382(6667):eadf1226, 2023. doi: 10.1126/science.adf1226.

[38] Dylan Kotliar, Adrian Veres, M Aurel Nagy, Shervin Tabrizi, Eran Hodis, Douglas A Melton, and Pardis C Sabeti. Identifying gene expression programs of cell-type identity and cellular activity with single-cell RNA-Seq. *eLife*, 8, 2019. doi: 10.7554/eLife.43803.

[39] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a prac-

tical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

[40] Vitalii Kleshchevnikov, Artem Shmatko, Emma Dann, Alexander Aivazidis, Hamish W King, Tong Li, Rasa Elmentaite, Artem Lomakin, Veronika Kedlian, Adam Gayoso, Mika Sarkin Jain, Jun Sung Park, Lauma Ramona, Elizabeth Tuck, Anna Arutyunyan, Roser Vento-Tormo, Moritz Gerstung, Louisa James, Oliver Stegle, and Omer Ali Bayraktar. Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature biotechnology*, 40(5):661–671, 2022. doi: 10.1038/s41587-021-01139-4.

[41] Nils Bessler, Amber K L Wezenaar, Hendrikus C R Ariese, Celina Honhoff, Noëlle Dommann, Ellen J Wehrens, Cristian Ruiz Moreno, Thijs J M van den Broek, Raphaël V U Collot, Daan J Kloosterman, Farid Keramati, Mieke Roosen, Sam de Blank, Esmée van Vliet, Mario Barrera Román, Lucrezia C D E Gatti, Ali Ertürk, Jürgen Kuball, Zsolt Sebestyén, Marcel Kool, Sara Patrizi, Evelina Miele, Annette Künkele, Mariëtte E G Kranendonk, Annelisa M Cornel, Stefan Nierkens, Christian Mayer, Hendrik G Stunnenberg, Anna Alemany, Maria Alieva, and Anne C Rios. De novo H3.3K27M-altered diffuse midline glioma in human brainstem organoids to dissect GD2 CAR T cell function. *Nature cancer*, 7(2):316–333, 2026. doi: 10.1038/s43018-025-01084-0.

[42] Shibla Abdulla, Brian Aevermann, Pedro Assis, Seve Badajoz, Sidney M Bell, Emanuele Bezzi, Batuhan Cakir, Jim Chaffer, Signe Chambers, J Michael Cherry, Tiffany Chi, Jennifer Chien, Leah Dorman, Pablo Garcia-Nieto, Nayib Gloria, Mim Hastie, Daniel Hegeman, Jason Hilton, Timmy Huang, Amanda Infeld, Ana-Maria Istrate, Ivana Jelic, Kuni Katsuya, Yang Joon Kim, Karen Liang, Mike Lin, Maximilian Lombardo, Bailey Marshall, Bruce Martin, Fran McDade, Colin Megill, Nikhil Patel, Alexander Predeus, Brian Raymor, Behnam Robatmili, Dave Rogers, Erica Rutherford, Dana Sadgat, Andrew Shin, Corinn Small, Trent Smith, Prathap Sridharan, Alexander Tarashansky, Norbert Tavares, Harley Thomas, Andrew Tolopko,

Meghan Urisko, Joyce Yan, Garabet Yeretssian, Jennifer Zamanian, Arathi Mani, Jonah Cool, and Ambrose Carr. CZ CELLxGENE Discover: a single-cell data platform for scalable exploration, analysis and modeling of aggregated data. *Nucleic acids research*, 53(D1):D886–D900, 2025. doi: 10.1093/nar/gkae1142.

[43] Samuel L Wolock, Romain Lopez, and Allon M Klein. Scrublet: Computational Identification of Cell Doublets in Single-Cell Transcriptomic Data. *Cell systems*, 8 (4):281–291.e9, 2019. doi: 10.1016/j.cels.2018.11.005.

[44] F Alexander Wolf, Philipp Angerer, and Fabian J Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome biology*, 19(1):15, 2018. doi: 10.1186/s13059-017-1382-0.

[45] Adam Frankish, Sílvia Carbonell-Sala, Mark Diekhans, Irwin Jungreis, Jane E Loveland, Jonathan M Mudge, Cristina Sisu, James C Wright, Carme Arnan, If Barnes, Abhimanyu Banerjee, Ruth Bennett, Andrew Berry, Alexandra Bignell, Carles Boix, Ferriol Calvet, Daniel Cerdán-Vélez, Fiona Cunningham, Claire Davidson, Sarah Donaldson, Cagatay Dursun, Reham Fatima, Stefano Giorgetti, Carlos Garcıa Giron, Jose Manuel Gonzalez, Matthew Hardy, Peter W Harrison, Thibaut Hourlier, Zoe Hollis, Toby Hunt, Benjamin James, Yunzhe Jiang, Rory Johnson, Mike Kay, Julien Lagarde, Fergal J Martin, Laura Martínez Gómez, Surag Nair, Pengyu Ni, Fernando Pozo, Vivek Ramalingam, Magali Ruffier, Bianca M Schmitt, Jacob M Schreiber, Emily Steed, Marie-Marthe Suner, Dulika Sumathipala, Irina Sycheva, Barbara Uszczynska-Ratajczak, Elizabeth Wass, Yucheng T Yang, Andrew Yates, Zahoor Zafrulla, Jyoti S Choudhary, Mark Gerstein, Roderic Guigo, Tim J P Hubbard, Manolis Kellis, Anshul Kundaje, Benedict Paten, Michael L Tress, and Paul Flicek. GENCODE: reference annotation for the human and mouse genomes in 2023. *Nucleic acids research*, 51(D1):D942–D949, 2023. doi: 10.1093/nar/gkac1071.

[46] Etienne Becht, Leland McInnes, John Healy, Charles-Antoine Dutertre, Immanuel W H Kwok, Lai Guan Ng, Florent Ginhoux, and Evan W Newell. Dimensionality reduction for visualizing single-cell data using UMAP. *Nature biotechnology*, 2018. doi: 10.1038/nbt.4314.

Author contributions

[To be completed]

Declaration of interests

[To be completed]

Declaration of generative AI and AI-assisted technologies in the writing process

[To be completed: state any use of generative AI in the preparation of this manuscript, per the journal's policy.]

Supplemental information

Supplemental information includes seven supplementary figures and one supplementary table, and can be found with this article online.

STAR⋆Methods

Key resources table

See Table 5.

Experimental model and subject participant details

This study reanalyses previously published human tumour and normal-brain data; no new human or animal material was collected. Every contributing study obtained its own ethical approval and consent, as stated in the primary publications listed in the key resources table. The reference cohort comprises 52 patients with H3K27M-altered diffuse

Table 5: Key resources table. Deposited data reused in this study and the software that produced the results. Version numbers are those recorded by `scripts/28_record_environment.py`; the full environment (557 conda and 167 pip entries for the local environment, plus the GPU image pins) is in `env/`.

Deposited data		
Dataset	Identifier	Source
Collot et al., DMG (snRNA)	Zenodo: 10.5281/zenodo.17922773 (scRNA.tar.gz)	Zenodo
DeSisto et al., paediatric HGG (scRNA)	GSE231859	GEO
Jessa et al., DMG (multiome RNA)	GSE210566	GEO
Jessa et al., DMG (sc/snRNA)	GSE210567	GEO
Okada et al., post-mortem DMG (snRNA)	EGAS50000001447 (raw, controlled); processed matrix from the study's UCSC Cell Browser instance	EGA / UCSC Cell Browser
Zang et al., DMG (scRNA)	GSE245674 (GSM7846769)	GEO
Braun et al., first-trimester human brain	CZ CELLxGENE Discover (collection 4d8fed08-2d6d-4692-b5ea-464f1d072077)	CZ CELLxGENE
Filbin et al., H3K27M glioma (Smart-seq2, TPM)	GSE102130	GEO
DMG immune cohort (SS2 + 10x, TPM)	GSE227983	GEO
Liu et al., H3-K27M DMG (Smart-seq2, TPM)	GSE184357	GEO
Siletti et al., adult human brain (pons dissections)	CZ CELLxGENE Discover (collection 283d65eb-dd53-496d-adb7-7570c7caa443)	CZ CELLxGENE
Whole-brainstem DIPG autopsy Visium	GSE280990	GEO
Collot et al., DMG Visium	Zenodo: 10.5281/zenodo.17922773 (ST.tar.gz)	Zenodo
Huchede et al., DMG Visium	GSE268577	GEO
Ren et al., DMG and glioblastoma Visium	GSE194329	GEO

Software and algorithms		
Software	Reference	Identifier
python v3.11.16	Python Software Foundation	`https://pypi.org/project/python/`
numpy v2.4.6	Harris et al.	`https://pypi.org/project/numpy/`
scipy v1.17.1	Virtanen et al.	`https://pypi.org/project/scipy/`
pandas v2.3.3	McKinney	`https://pypi.org/project/pandas/`
anndata v0.12.19	Virshup et al.	`https://pypi.org/project/anndata/`
scanpy v1.11.5	Wolf et al.	`https://pypi.org/project/scanpy/`
scikit-learn v1.7.2	Pedregosa et al.	`https://pypi.org/project/scikit-learn/`
statsmodels v0.15.0	Seabold and Perktold	`https://pypi.org/project/statsmodels/`
matplotlib v3.11.2	Hunter	`https://pypi.org/project/matplotlib/`
infercnvpy v0.6.1	Sturm et al. (inferCNV port)	`https://pypi.org/project/infercnvpy/`
leidenalg v0.12.0	Traag et al.	`https://pypi.org/project/leidenalg/`
umap-learn v0.5.12	McInnes et al.	`https://pypi.org/project/umap-learn/`
scvi-tools v1.4.2	Lopez et al.; Xu et al.	`https://pypi.org/project/scvi-tools/`
torch v2.14.1	Paszke et al.	`https://pypi.org/project/torch/`
cell2location v==0.1.5	Kleshchevnikov et al.	`https://pypi.org/project/cell2location/`
pyro-ppl v1.9.1+ab0491a	Bingham et al.	`https://pypi.org/project/pyro-ppl/`
lifelines v0.30.3	Davidson-Pilon	`https://pypi.org/project/lifelines/`

midline glioma (42 pontine and 15 thalamic samples); age group, anatomical site, assay and reported H3 variant per sample are given in Table 1. No mouse data were used.

Method details

Cohort assembly

Public deposits were screened by GEO E-utilities queries and a scholarly-literature search for human H3K27M DMG single-cell and spatial data (Supplementary Table 1). Six cohorts with recoverable UMI counts form the reference (GEO GSE210567 and GSE210566[16]; Zenodo 10.5281/zenodo.17922773[23]; GEO GSE231859[27]; GEO GSE245674, sample GSM7846769 only[33]; and the post-mortem series of Okada et al.[26], raw data under EGA accession EGAS50000001447 with the processed matrix obtained from its UCSC Cell Browser instance). Five cohorts are projection-only: GEO GSE102130[15] and GSE184357[18], both deposited as transcripts per million; GSE227983; and the normal-brain references of Braun et al.[37] (hindbrain, pons and medulla dissections, 286,973 cells) and Siletti et al.[36] (six pons dissections, 204,168 nuclei), both obtained from CZ CELLxGENE Discover[42]. Spatial cohorts are GSE268577[25], GSE194329[24], GSE280990 and the spatial archive of Zenodo 10.5281/zenodo.17922773. One cohort (ScPCA SCPCP000021) was excluded after quality control. All data are human; no mouse data were used.

Quality control

Thresholds were applied per assay class with distinct mitochondrial-fraction limits for whole cells and nuclei, a minimum gene count per cell, and doublet scoring by Scrublet[43] at a fixed threshold of 0.25. Samples failing a per-sample gate were flagged and excluded from the reference but retained in the released object with their flag. Of 420,381 cells in atlas-eligible samples, 356,406 passed; the 311,618-cell reference is the UMI-count subset of those. All processing used Scanpy[44].

Malignant-cell calling

For each sample, cells were scored against curated lineage marker sets, and an inferCNV-style profile was computed from log-normalised expression relative to within-sample immune and oligodendrocyte references over 100-gene windows ordered by GENCODE v32 coordinates[45], summarised both chromosome-wide and per arm. Cluster-level calls combined an aggregate score threshold with recurrent arm-level events (1q, 2, 7 gain; 10q, 13q, 14q, 17p loss). A three-tier consensus — marker only, copy number only, and the agreement of both — was evaluated against author annotations per sample; the released `consensus_call` requires marker and copy-number agreement, and `consensus_call_eventtrust` is the permissive sensitivity variant.

Integration and label transfer

scVI[29] was trained on 3,000 highly variable genes (Seurat v3 selection with per-sample batch[34]) with sample as batch key, study, assay and mitochondrial fraction as covariates, 30 latent dimensions and three random seeds, on a single A10G GPU. Embeddings were compared with scIB-style metrics[32] and selected by the pre-stated 0.4/0.6 batch/biology score; UMAP was used for visualisation only[46]. scANVI[30] was initialised from the chosen scVI model and trained on all labelled cells with the validation split stratified by patient. Query cohorts were encoded by the frozen reference encoder without architectural surgery or fine-tuning, which the released run metadata records as zero training epochs; labels were transferred both by the scANVI classifier and by 15-nearest-neighbour vote in latent space[31], and both are released per cell with a neighbour-distance uncertainty percentile.

Metaprograms

Per-sample consensus non-negative matrix factorisation[38] was run on malignant cells from samples with at least 200 such cells, with k chosen per sample by stability and capped by cell number. Programmes were clustered by Jaccard similarity of their top genes; the primary dendrogram cut and the pre-stated replication rule (at least three studies and both assay classes) were fixed before inspection, and the sensitivity of both the metaprogram

count and the replication count to the cut was recorded. Per-cell family assignment used within-sample standardised programme scores. The assay-masking control removed the 6,013 genes called assay-biased from the two paired multiome samples (5,921 by a two-sample Welch test with consistent sign in both pairs, plus 92 mitochondrial and ribosomal genes) and was compared against ten size-matched random masks.

Spatial deconvolution and statistics

cell2location[40] reference signatures were estimated from the 13-class metaprogram-family reference (250 epochs), and each section was deconvolved with a wall-clock-budgeted schedule up to 20,000 epochs; realised epoch counts and full ELBO histories are released, and convergence is argued from those histories rather than from reaching the target. Co-localisation was computed on spot proportions (with a centred-log-ratio variant as sensitivity), Fisher-z averaged within patient, pooled by DerSimonian–Laird random effects, and corrected across 42 state-by-microenvironment pairs[39]. Serial sections of one autopsy count as one patient, as do the core and edge sections of one resection.

Audit

The completed analysis was audited by six independent adversarial reviews with no access to the reasoning that produced it, yielding 163 findings (4 critical, 33 major). Findings were remediated in four waves and the closure of every critical and major finding was then verified by an independent re-audit that recomputed each quantity from the released files: 78 of 90 findings in scope verified closed, including 4 of 4 critical. The complete record — findings, closure claims, recomputed values and remaining open items — is released with the atlas.

Quantification and statistical analysis

All analysis ran on a 14-core laptop with 24 GB of memory except scVI/scANVI training and cell2location deconvolution, which ran on a single A10G GPU. Software versions are recorded for both environments. Bibliographic records were retrieved from PubMed

(National Center for Biotechnology Information); DOIs for all cited works are given in the reference list.

Supplementary information

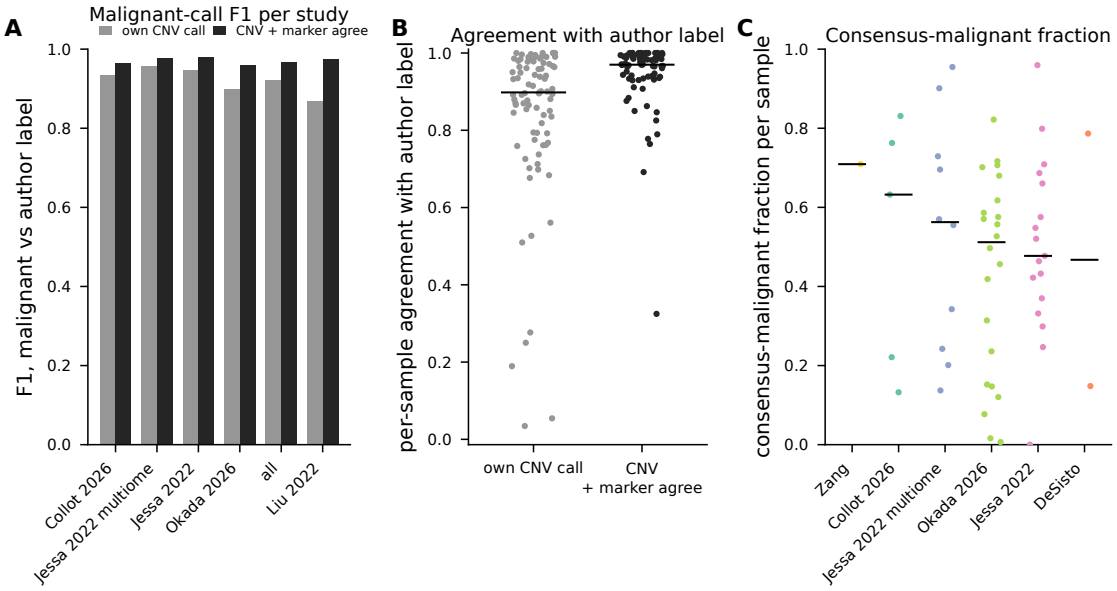


Figure S1: Malignant-cell calls. **a**, F1 of the author-independent malignant calls against author labels per study: own copy-number call, and cells where the copy-number and marker calls agree. **b**, Per-sample agreement with author labels (samples with at least 100 compared cells). **c**, Fraction of consensus-malignant cells per reference sample, by study.

Table S1: Supplementary Table 1 | Public deposits screened and their role in this study. Accession, repository, access status and deposited cell or spot count for every deposit screened, with an explicit column for whether it entered the released results. The final column is the role proposed at screening, which is not the inclusion record: deposits marked "no" were dropped before or during quality control (the mouse developmental reference and the adult Paired-Tag reference were never used, and the ScPCA samples failed quality control), and the Phase 7 bulk cohort is outside the scope of this paper.

Dataset	Accession	Repository	Access	Cells/spots	Used here	Role at screening
Jessa_2022_scRNA_snRNA	GSE210567	GEO	open	176,912 QC-…	yes	CORE integrated (scVI/…
Jessa_2022_multiome	GSE210566	GEO	open	not stated …	yes	CORE integrated + chro…
Collot_2026_scRNA_snRNA (s…	10.5281/zenodo.17922773	Zenodo	open	73,579	yes	CORE integrated
DeSisto_2024	GSE231859	GEO	open	15,617	yes	CORE integrated (after…
Filbin_2018	GSE102130	GEO	open	4,039 cells	yes	PROJECTION (scANVI que…
Liu_2022	GSE184357	GEO	open	9,248 post-…	yes	PROJECTION (or integra…
LaBelle_2025	GSE227983 (+GSE271936, 4 e…	GEO	open (processed …	30,892	yes	PROJECTION (TPM only, …
Huchede_2024	GSE268577	GEO	open	1,618 spots	yes	SPATIAL validation
Ren_2023_DMG_Visium	GSE194329 (5 DMG + 5 GBM +…	GEO	open	not stated …	yes	SPATIAL validation
Collot_2026_Visium (S…	10.5281/zenodo.17922773	Zenodo	open	not stated	yes	SPATIAL validation
Xie_2026_adult_brain_paired-Tag	GSE274407	GEO	open	36,646 pons…	no	NORMAL reference (adul…
Siletti_2023_adult_human_brain C…	10.1126/science.add7046	gene	open	pons subset…	yes	NORMAL reference (adul…
Braun_2023_first_trimester_brain C…	10.1126/science.adf1226	gene / Li…	open	hindbrain s…	yes	NORMAL reference (deve…
OpenPedCan_PBTA_DMG	PBTA/OpenPedCan v14	D3b open-acces…	open	nan	no	BULK survival validati…
Zang_2024_DMG_sample2	GSE245674 (GSM7846769 only…	GEO	open	half of 10…	yes	CORE integrated (minor…
ScPCA_SCPCP000021_HOPE	SCPCS000021 (SCPCS000636, …	ScPCA portal (…	open (accept por…	3,985 (1,05…	no	CORE integrated (minor…
Okada_2026_postmortem_brain	EGAS50000001447 (raw); UCS…	EGA (controlle…	raw controlled; …	not stated	yes	CONDITIONAL integrated…
Whole_brainstem_DIPG_2025	GSE280250	GEO	open	not stated	yes	SPATIAL validation (in…
Jessa_mouse_developing_pons_5	GSE188635 (su…	GEO	open	not stated	no	NORMAL reference (mous…
ScPCA_SCPCP000001_HGG_midline	SCPCS000001 (HGG midline-loc…	ScPCA portal	open (accept por…	12,000 acr…	no	CONDITIONAL integrated
Sussman_2026_longitudinal_pHGG (HTAN) [UN…	dbGaP phs002371	dbGaP	controlled	nan	no	WATCH (controlled acce…
Stanford_DIPG_CART_autopsy_sn…	Abstract DMG-48, Neuro-Onc…	none yet	not released	320,000	no	WATCH (unreleased)
Acta_Neuropathol_2026_DIPG_scR…	EGA 'Upon request' (not ye…	EGA	on request	nan	no	EXCLUDE
Cell-line_PDX_series	GSE212782, GSE268775/6, GS…	GEO	open	nan	no	EXCLUDE (models / in v…
CSF_and_PBMC_immune_atlases	GSE261235, GSE186802, GSE2…	GEO	open	nan	no	EXCLUDE (not tumour ti…

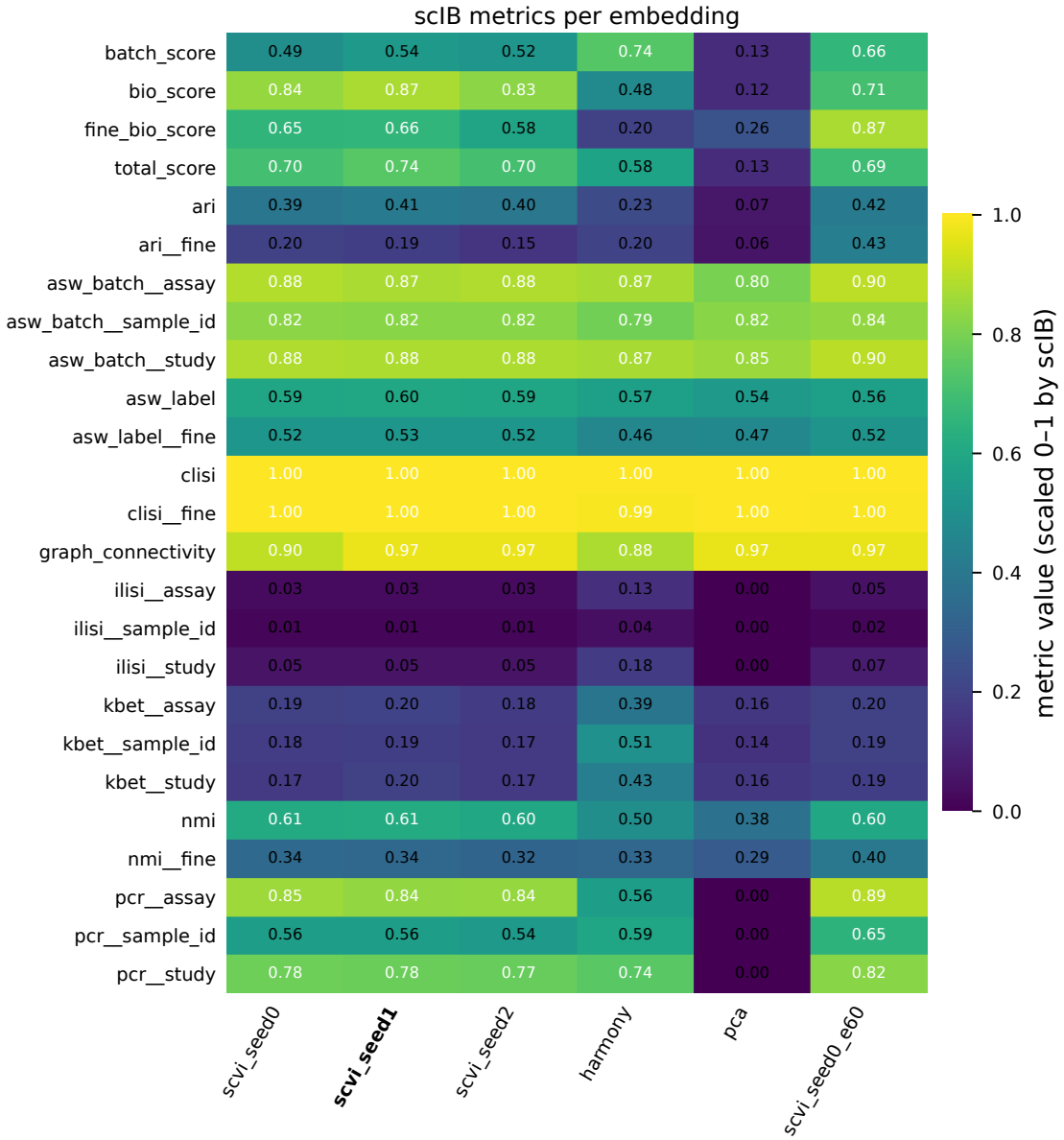


Figure S2: Integration benchmark. All batch-correction metrics (batch keys sample, study, assay) and biological-conservation metrics (coarse and fine labels) for the six candidate embeddings, with summary scores above. Assay remains the hardest batch key for the chosen model.

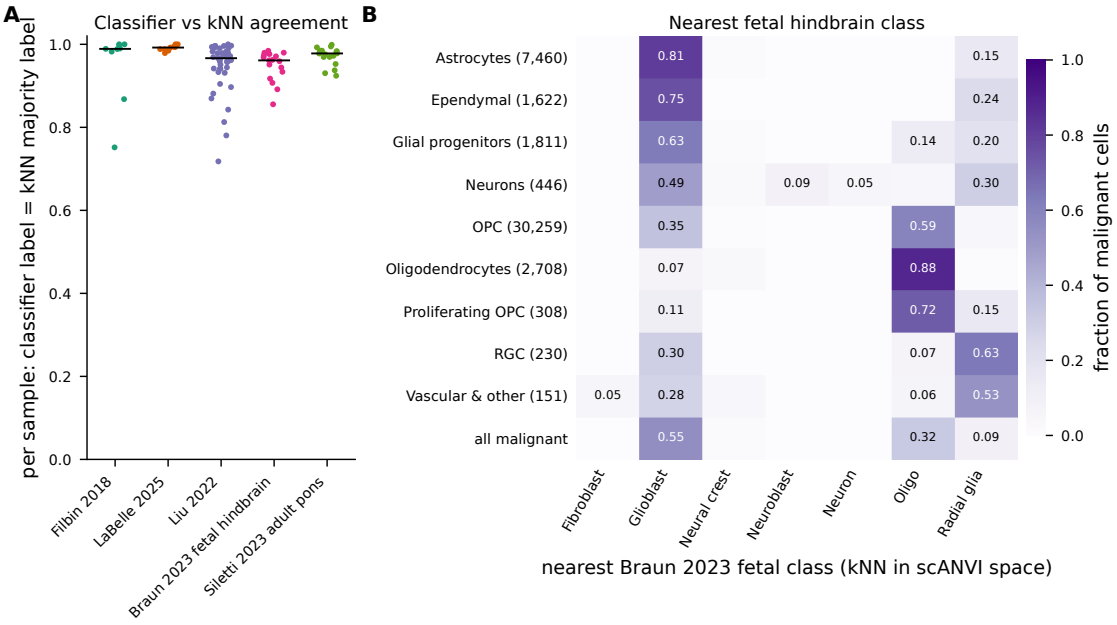


Figure S3: Mapping confidence and nearest fetal class. a, Per-sample fraction of query cells whose classifier label equals the majority label of their 15 nearest reference cells. **b**, Nearest fetal hindbrain class of malignant cells, overall and by author state.

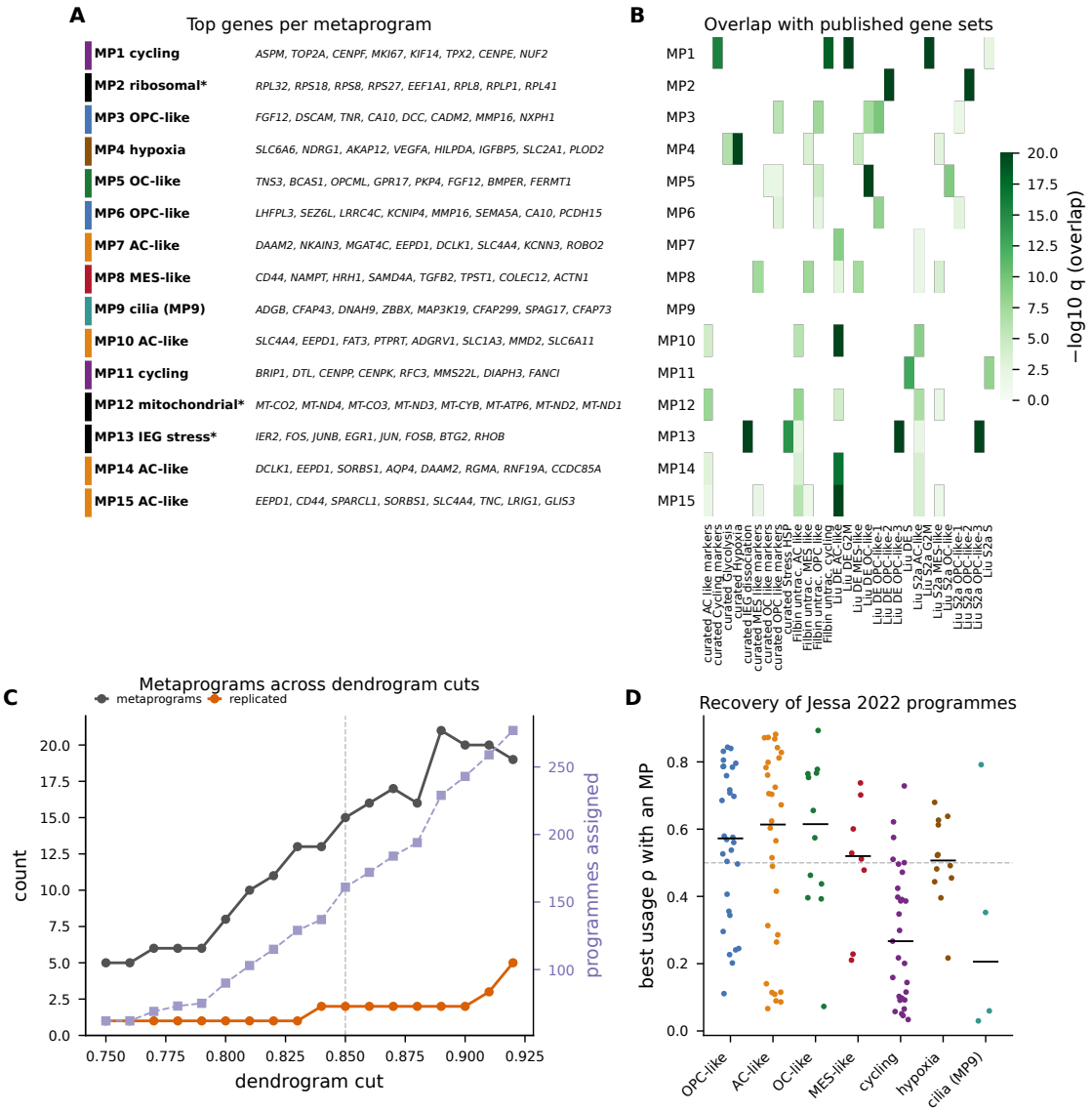


Figure S4: Metaprogram content and robustness. **a**, Top eight genes of each metaprogram; a trailing asterisk marks a technical programme. **b**, Overlap of metaprogram top-50 genes with published DMG and glioma state gene sets (hypergeometric test, Benjamini–Hochberg q). **c**, Number of metaprograms, replicated metaprograms and assigned programmes across dendrogram cuts. **d**, Best within-sample correlation of published per-sample programmes with an atlas metaprogram.

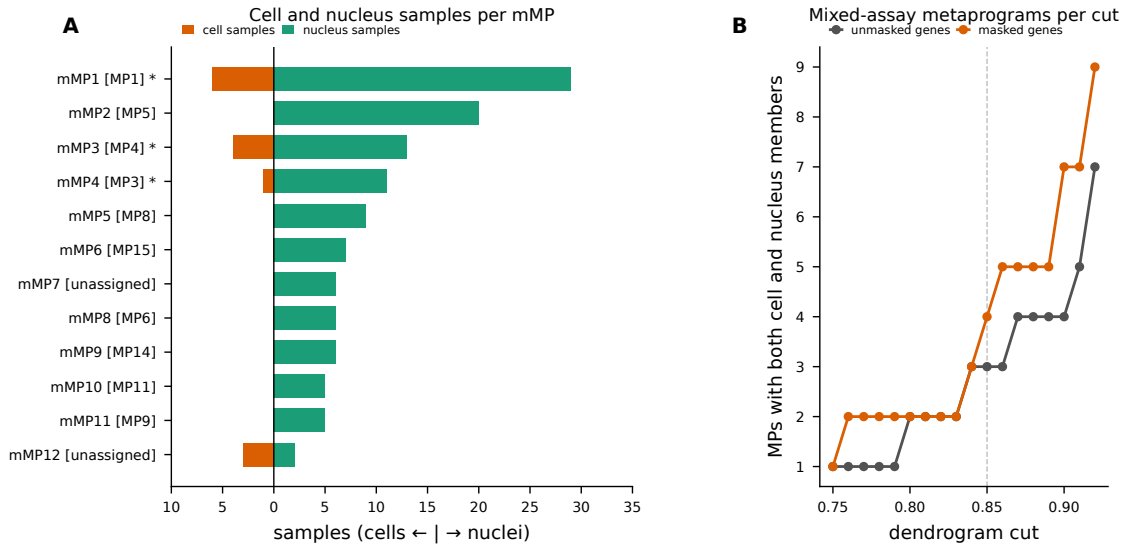


Figure S5: Metaprograms after masking assay-biased genes. **a**, The 12 masked metaprograms at the primary cut, with contributing cell and nucleus samples. **b**, Metaprograms containing both cell- and nucleus-derived programmes across cuts, with all genes and after masking.

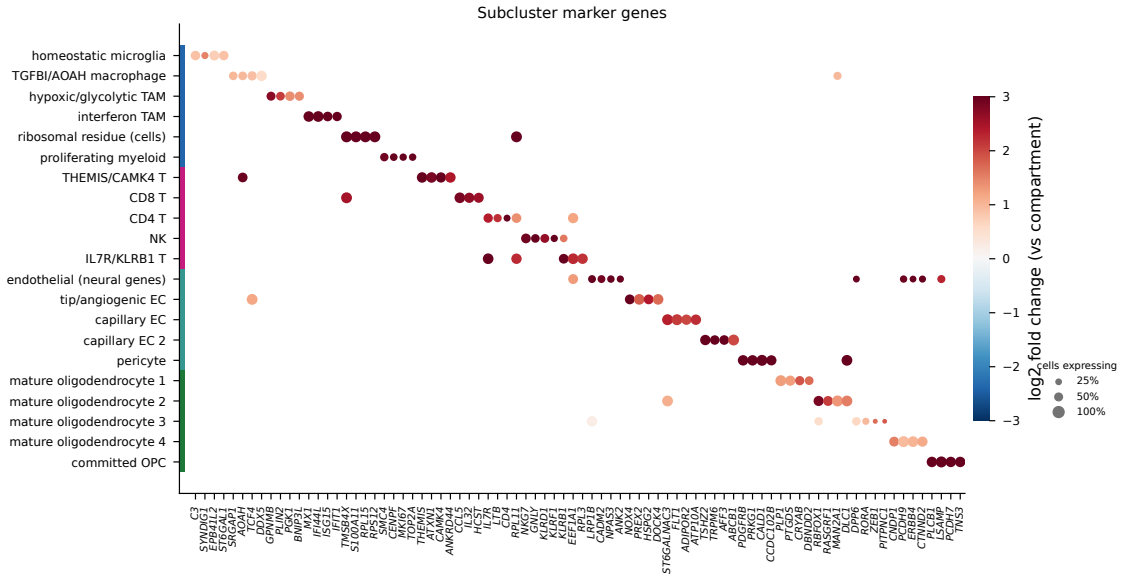


Figure S6: Marker genes of the microenvironment subclusters. Top four genes per subcluster; dot size is the fraction of the subcluster's cells expressing the gene and colour the log fold change against the other subclusters of the same compartment.

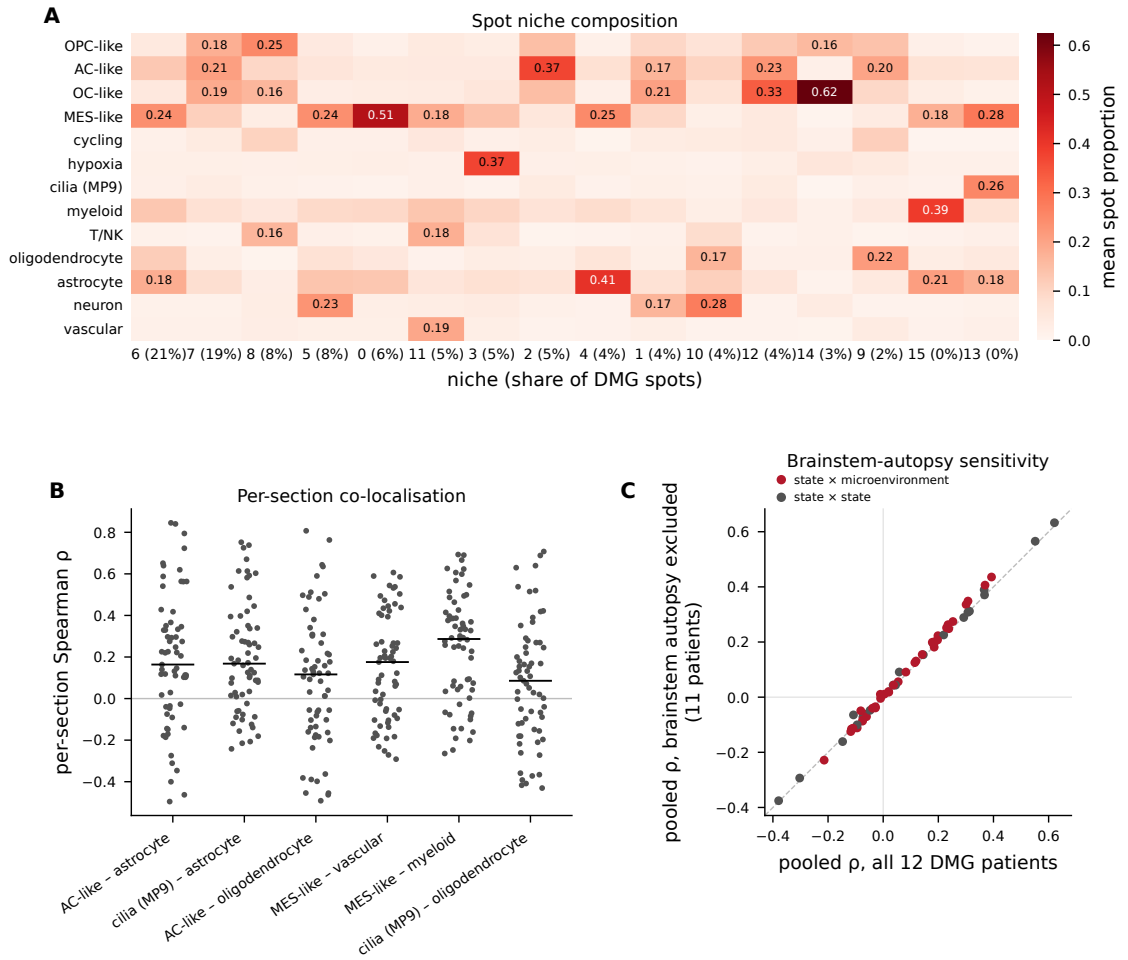


Figure S7: Spatial niches and sensitivity analyses. **a**, Mean class proportions of the 16 spot niches. **b**, Per-section correlations for the six strongest pairs of Figure 5d; every point is a section, ten of which are serial sections of one autopsy. **c**, Pooled correlations with and without that autopsy.